UNIT 2 – DIGITAL MAPPING CONCEPT

- Map concept: map elements, map layers, map scales and representation
- Map projection: coordinate system and projection system

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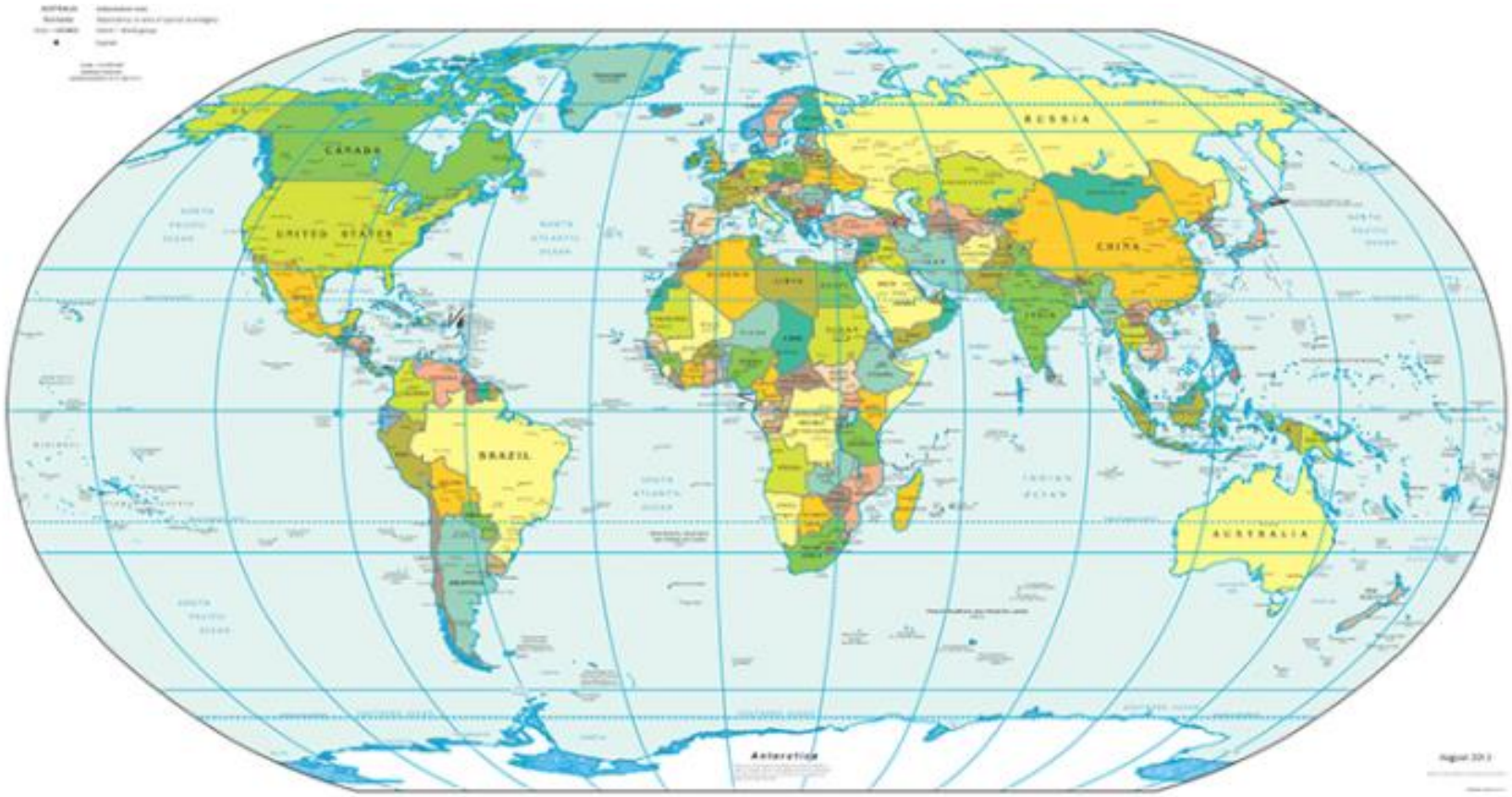
Map



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What is Map???????

- A map is defined as a representation, usually on a flat surface, of a whole or part of an area.
- The job of a map is to describe spatial relationships of specific features that the map aims to represent.
- There are many different types of maps that attempt to represent specific things.
- Maps can display political boundaries, population, physical features, natural resources, roads, climates, elevation (topography), and economic activities.

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- Maps are produced by cartographers. Cartography refers both the study of maps and the process of map-making. It has evolved from basic drawings of maps to the use of computers and other technologies to assist in making and mass producing maps.

There are four basic kinds of maps

Physical

Transportation

Political

Historical

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IS A GLOBE A MAP?

A globe is a map. Globes are some of the most accurate maps that exist.

This is because the earth is a three-dimensional object that is close to spherical.

A globe is an accurate representation of the spherical shape of the world.

Maps lose their accuracy because they are actually **projections** of a part of or the entire Earth.

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Elements of Maps

1. **Map Has a Clear Purpose**

- What is the map intention? Are you achieving it? Designing the concept of a map is typically how cartographers begin. Ideas are driven by purpose.
- Maps help people understand their place on Earth. Whether you're mapping geology, census data or watersheds, your map's purpose helps readers spatially understand a theme or subject.

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Map Titles Reflect the Author's Intention

- The map title should reflect the purpose of the map. The title is usually based on the information being displayed in your map. It usually includes your theme and geographic location.
- Your title shouldn't leave your readers guessing its purpose. The map title reveals important information to the reader. It's the intention worded to the map content.

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- **Use the Appropriate Template**
- Some companies and organizations have map templates on hand. Throw in the data. Change a couple of map elements. With a few clicks of the mouse, there you have it! Map making made easy.
- When selecting the template, keep the map orientation in mind. There is landscape or portrait orientation and different map sizes. If map features are linear and run east-west (like pipelines or highways) use a horizontal map layout.

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- **Add Company and Client Logo**
- Chances are your map will be designed for someone. Adding your organization and the client's logo on the map is a way to recognize the author and who the map product was designed for.
- Did the client (if there is one) ask to have its logo on the map? Are you using the newest one? When in doubt, seek permission to use somebody else's logos or name,

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Ensure Correct Map Extent and Coverage

- Maps help us view the world in a scaled down fashion. But how scaled down should your map be? Understanding the scale of your map features is one of the first steps in map design.
- Is your data the scale of a continent or an airport. Large and small-scale maps are expressed in ratios. They're usually rounded such as 1:100. Pan to the center so your main feature is front-and-center. Alternatively, you should ensure other important features are included as a reference. The map making process must always consider extent and coverage.

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Define a Projection

- The Earth is curved. Map projections flatten our curved Earth and always adds distortion. They transform the sphere on a plane. Cartographers display maps in two dimensions.
- Whether you want to preserve area, direction or scale, choosing your map projection is not always an easy task. After defining your projection, list the projection in your map's metadata.

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Display the Correct Content

- You cringe at the thought of using incorrect data in your maps. GIS technicians revise data until it’s deemed final. In the end, you have to ask yourself – am I linking to the correct file?
- Maps are abstract representations of Earth. But the correctness relies in the accuracy of the data that’s being represented. Do the extra detective work to ensure the correct data is being shown.

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Give Perspective with an Inset Map

- It’s one of the essential map elements. A well-designed map often includes a key map. A key map puts your geography into perspective at a smaller scale. It’s a smaller, separate map that shows the extent of the map in relation to a larger area.
- But they’re also used to clarify features on a map. Inset maps can focus in on areas of interest. They enlarge features in smaller maps giving readers more perspective.

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Key Map – a Map Within a Map, Within a Map

- A map within a map, within a map. Confused yet? These types of maps are tremendous assets for mammoth-sized projects. It gives even more perspective to the reader and communicates ideas even more effectively.
- During the construction of a new pipeline spanning an entire continent, the main map may show a water crossing. The inset map may show the state or county. Another map at an even smaller scale (zoomed out) can show all the water crossings in the United States.

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Decide Your Labeling/Annotation

- Turn nothing into something. Labels need to be placed with purpose. But busy maps with too much labeling is confusing with no better place than a trash bin.
- The general rule is that labels go left-right and south-north. Font face, color and size need to be considered. Labels can follow the curvature in features. They shouldn’t overlap lines because they become unreadable. It’s good practice to offset preferably on top. Annotation font size can be scaled larger for more important features.

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- **Glue Readers in with Neat Lines**
- Neat lines are good at keeping readers glued to your maps. Readers eyeballs become locked in because neat lines wrap all your content with one solid border.
- There not always a dire need for neat lines. But a lot of the best map makers use neat lines because the surrounding border just belongs on a map as a container.

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Order Layers by Importance

- Data layers need to be ordered properly for map making. The correct hierarchy of layers in a map determines whether or not features will be on top or below others.
- If you have fish observations under an ocean polygon, they will completely disappear in the map. Having a logical order of map layers ensures the correct number of features will be displayed.

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- **View All Ingredients in Your Map with Legends**
- Maps are abstractions of reality. They use map symbols to generalize our complex world we live in. Legends tell the map reader what the polygons, lines, points or grid cells represents. Map legends explain to the reader what these features represent. They are one of the most important map elements to consider during map making.
- Readers can immediately learn everything in your map from start to finish like a list of ingredients in a recipe. Keep your legend clean with items aligned to the left. Remove unnecessary items that aren’t in your maps. Generally, legends follow the order points, lines and polygons.

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Measure Length with Scale Bars

- A scale bar is a map element that graphically shows the distance of units on a map. They are another effective way to gain a better understanding of scale.
- Readers like to glance at scale bars. They need a way to scale feature on a map as a way to measure distance. Scale bars used in map making are practical and generate instant answers for size and measurements.

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Know the Date of Production

- In crunch time, it’s important to understand the date of production. Was the map created this decade? Is the information in the map still relevant?
- Things are always changing in our dynamic world. Dates of production eliminates future confusion. It can hit you like a ton of bricks. It’s surprising how often map production dates are missing.

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Metadata is Your Map's Narrator

- Cartographers use maps to tell a story. Metadata is like having a narrator tell readers about the story. It's a summary of the data and information about the map.
- Metadata is not always the most exciting topic in GIS. But knowing the origin of your map data is insanely important. Include metadata about the spatial data and all of a sudden you've given readers a narration of your map.

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Orientate Yourself with a North Arrow

- Just like your standard compass, north arrows gives the orientation of north direction. North arrows are one of the essential map elements you see on almost every map. Not to be confused with the magnetic north – **our dynamic planet's magnetic north is always changing**.
- North arrows point in the geographic north cardinal direction. With little effort, readers can orientate themselves northerly. Map making has to consider north orientation.

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- **Fill in the Surrounding Map Data**
- When you show ortho photography in a map, it’s sometimes clipped to limited extent. This results in a big void of white or black in your map. This disrupts your map’s harmony. It attracts attention to these voided areas.
- Although it’s not always necessary, it’s prudent to fill in this void. Google, Bing and OpenStreetMap have free services that provide basemap imagery (with attribution) to fill in the voids.

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Cover Yourself with the Appropriate Disclaimer

- Disclaimers address the legal repercussions of mapping and liability. It’s not exactly a cartographer’s dream adding in their maps. But sometimes disclaimers may be what saves you in the end.
- Disclaimers can pay dividends. They are a glimmer of hope if all goes wrong in your map. Although extremely rare, disclaimers give you a piece of mind. Get your standard disclaimer checked by a legal representative before adding it to a map.

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Fill Out Data Sources Information

- It’s easy to cut corners and forget to list your maps data sources. Because to be honest, not really a lot of people tend to look at them.
- Just like a thesis paper, maps can include references. During map making, you can insert footnotes with a list of sources including author and year. When someone looks at your sources, you have your bases covered.

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- **Add Graticules as Handy Guiding Lines**
- Graticules represent meridians or parallels (or lines in other projections). They are lines on your map running north-south and east-west. Measured grids are for projected coordinates.
- Although graticules are not always necessary in a map, they can be handy guides for horizontal and vertical measurements. Follow a graticule and instantly understand latitude or longitude position. A map reader can estimate a features coordinates by looking at graticules in cartographic maps.

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Ensure Standard Symbology is Used

- Symbology is the cartographic language that is communicated to the audience. It’s the graphic representation of real world features on a map. We learn the maps story through well-chosen symbology.
- One of the main goals of cartography is to ensure individuals reviewing the map can understand its significance. You instantly lose the confidence of the map reader with inconsistent symbols

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Use Appropriate Scale Symbology

- Symbols on a map are like words in a book. They have to be intuitive. They describe information to the reader. And symbology has to use the appropriate scale
- Some symbology is universal. Cartographers are concerned how symbols are perceived on a map. Symbols are sometimes exaggerated in map making so readers can quickly understand. A small-scale map could have a city represented as a point. Zoom into a large-scale map and the city is better displayed as a polygon boundary.

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Select Colors that Reflect a Theme and Purpose

- Color catches your eye. It's usually the first place you look in a map. Your eyes will beam in on color. Color is used to clarify features. It increases design possibilities. Maps reveal information on location and attributes. Combined with color, it depicts relationships and adds visual clarity.
- People respond to color differently. Blue is generally water and positivity. Green is often used for vegetation. Brown is often used for mountains and dry land. Red often represents important features. **ColorBrewer** is a helpful resource to choose meaningful colors in your map making process.

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Think Outside the Box with Different Map Types

- Poorly designed maps are difficult to understand. For example, having a thematic map with too many color shades easily frustrates readers. The idea of the map won’t be conveyed.
- A choropleth map is good for population data. A climate map is good for showing temperature change. Bar graphs are good at showing change of numerical values. Cartographers have plenty of tools in their toolbox for selecting the right map type.

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Carefully Select Typeface

- The Comic Sans font has become the laughing-stock of typeface design. There's no place on a map for this font. Legibility is always a concern whether you use a serif or sans-serif font.
- A general rule is that base features use Times New Roman and thematic features use Arial. But it depends on the map design. Is it a formal, informal, classic or contemporary style map? **TypeBrewer** is a useful resource that assists you in selecting typography in your cartographic maps.

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Limit the Number of Features

- When you have too many features in your map, it quickly becomes a horrible game of “Where’s Waldo”. Because in the end, you are trying to communicate that one map purpose. As soon as you add unnecessary features, your map may be deemed a cartographic nightmare.
- With too many features, information isn’t passed to the reader. It gets miscommunicated. When it’s all said and done, you may be better off splitting a map into two. Having too many features complicates your map.

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Italicize Water Features

- It’s a kind of rule that water is always in italics and its blue. You rarely see it in other formats. When you do, they seem out-of-place.
- Rivers curve and so should your text. Slanted text shows fluidity in text. Scale text for larger lakes. Decrease text size for smaller and less important lakes. Map making has to consider font face and style when labeling different features.

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- **Beware of Overlapping Text**
- Label placement is surprisingly one of the biggest challenges cartographers face. Readability is the ultimate goal. The key is to add contrast from the background color and the label. This ensures that your text will be readable. Halos around text can improve visibility.
- With many features labelled in a small area, it can quickly become a cartographic nightmare. Labels can easily clash with one another. One option is to place text manually to avoid conflicts. Call-outs can also improve visibility.

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Carve Out the Author’s and Reviewer’s Name to the Map

- Who made the map? Who reviewed it? Not necessary in all map making design but a quick fix is to add the cartographer and reviewer names to the map.
- During questions and answer time, it makes all the difference. Have pride in your work.

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Run a Spell Check

- Maps always reveal positions. But they also reveal textual information. Place names, legend items and metadata are all in text.
- You instantly lose respect from your audience with incorrect spelling. This simple task makes or breaks it. Check your spelling on the map and in the legend.

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Features in Legend are Present and Singular

- You learn from maps. You extract information from maps. But when a feature in your map isn’t labeled or in the legend, information is lost.
- It’s good practice to manually go through entities on a map and your items listed on the legend. It’s been a habit that legend items are singular for map making.

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Find Balance & Harmony

- You learn about map balance and harmony with experience. It’s not something you pick up after your first map.
- When you learn to organize map and other elements, you start figuring out balance. Achieving harmony is the impression of equilibrium. Well-balanced cartographic design is formed with visual weight and visual direction.

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Maps are generally accepted as precise and accurate, which is true but only to a point.
A map of the entire world, without distortion of any kind, has yet to be produced; therefore it is vital that one questions where that distortion is on the map that they are using.

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Map Layer

- Maps in ArcGIS are composed of a series of map layers drawn in a particular order. A map layer defines how a GIS dataset is symbolized and labeled (that is, portrayed) in your map views.
- **Maps** are built by selecting a basemap and adding **layers** from ArcGIS Online, the web, **map** notes, or files on your computer.
- A layer represents geographic data in ArcMap, such as a particular theme of data. Examples of map layers include streams and lakes, terrain, roads, political boundaries, parcels, building footprints, utility lines, and orthophoto imagery

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Layers are also the way that you work with GIS datasets in ArcGIS, for example:

- Layers define how features identify and report themselves.
- Layers are used to edit geodatabase datasets.
- Layers are used to define how you work with feature attributes.
- Layers can be saved to a file on disk as a layer file (.lyr). A layer can also be saved with its data as a layer package (.lpk). When you add a layer file to another map, it will draw exactly as it was saved.
- You can share layers and layer packages over the network, on the web, and via e-mail.

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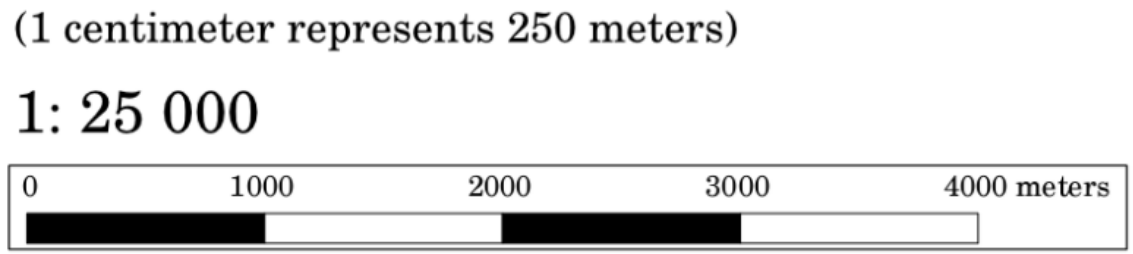
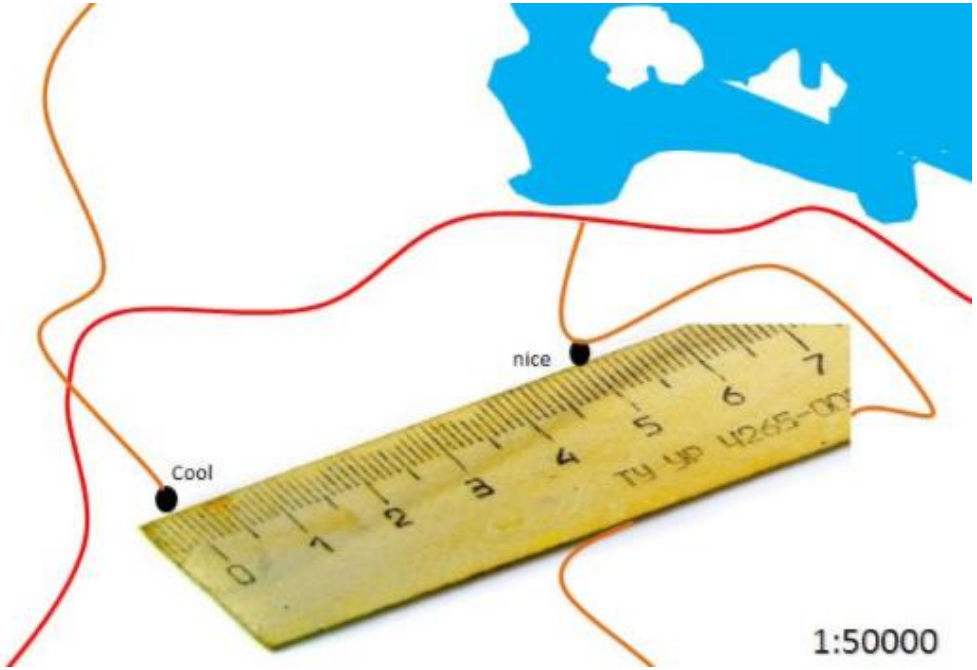
Map Scale



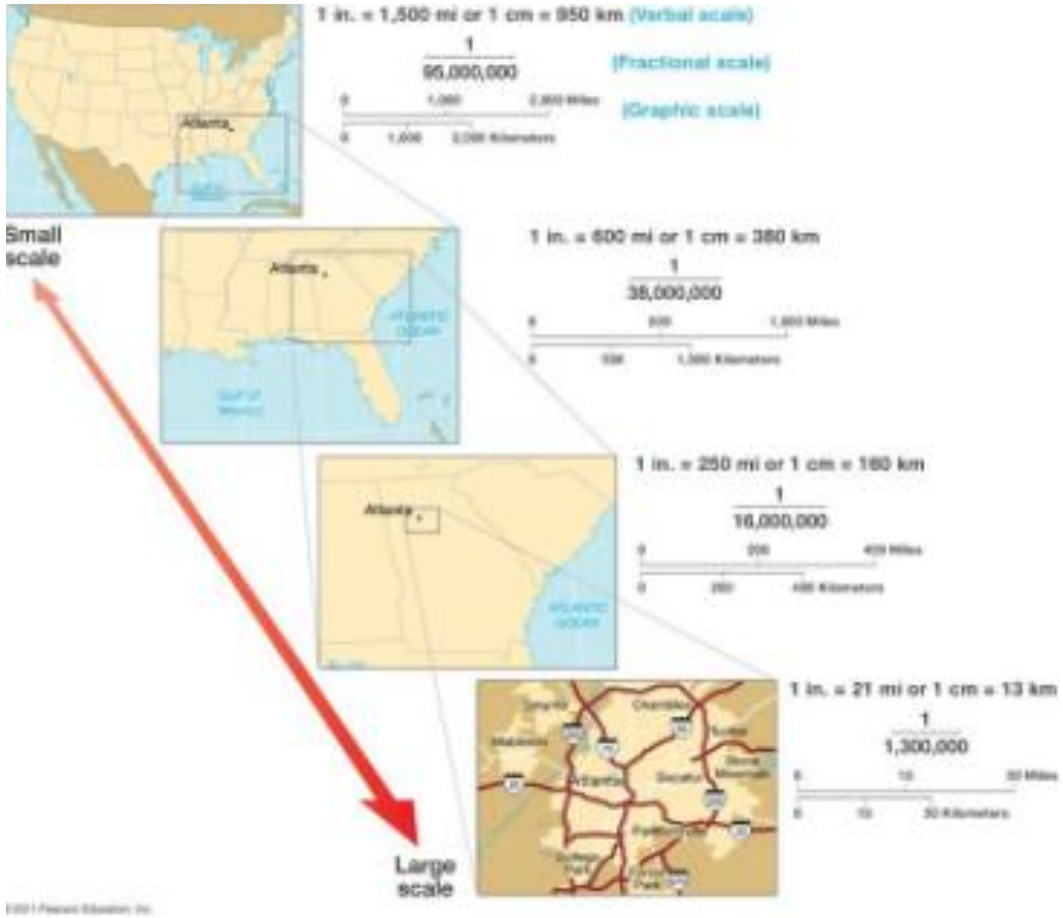
- **Map scale** refers to the relationship (or ratio) between distance on a **map** and the corresponding distance on the ground.
- For example, on a 1:100000 **scale map**,

1cm on the **map** equals 1km on the ground.

- Scales are used to find the actual distances between two points on map. They are used to get a better idea of the actual size of an area .



Large and Small Scale



Small-scale maps depict larger territories, but large-scale maps present more detail.

1:5000 is a large scale that shows small area with more details and 1:50000 is a small scale that depicts large area with less detail

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Map Projection

- There are several types of map projections, as well as several methods used to achieve these projections.
- Each projection is most accurate at its center point and becomes more distorted the further away from the center that it gets.
- The projections are generally named after either the person who first used it, the method used to produce it, or a combination of the two.

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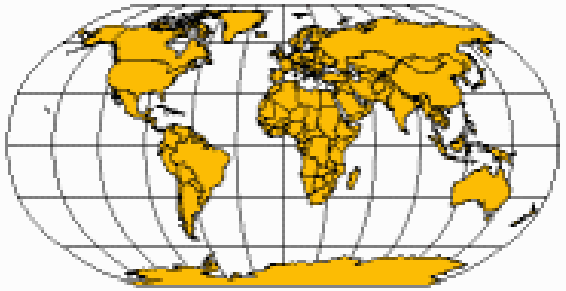
Map Projection

Flat Map

Cartesian coordinates: x,y

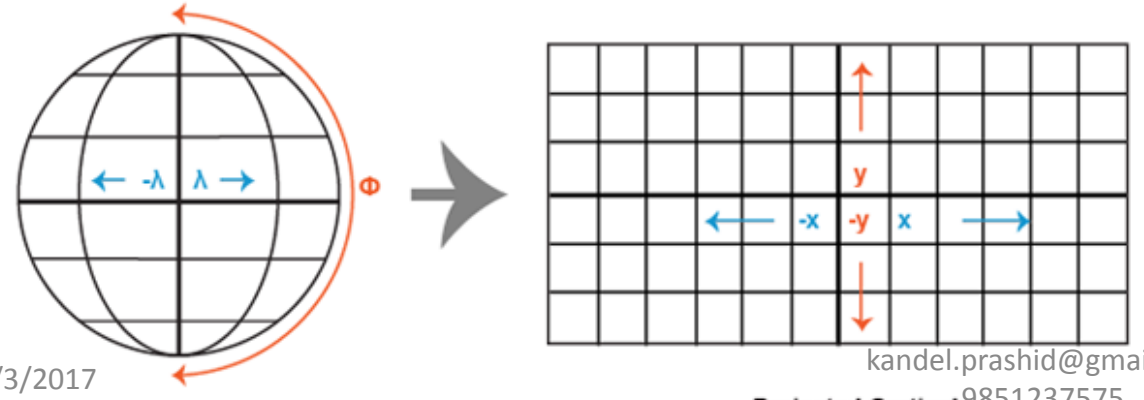
(Easting & Northing)



Curved Earth

Geographic coordinates: ϕ, λ

(Latitude & Longitude)



Graticule on sphere

Projected Graticule

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UNIT 3 – SPATIAL DATA MODELING AND DATABASE DESIGN

- Introduction to geographic phenomena and data modeling
- Spatial relationships and topology
- Scale and resolution
- Vector, raster and digital terrain model
- Spatial database design with the concepts of geodatabase.

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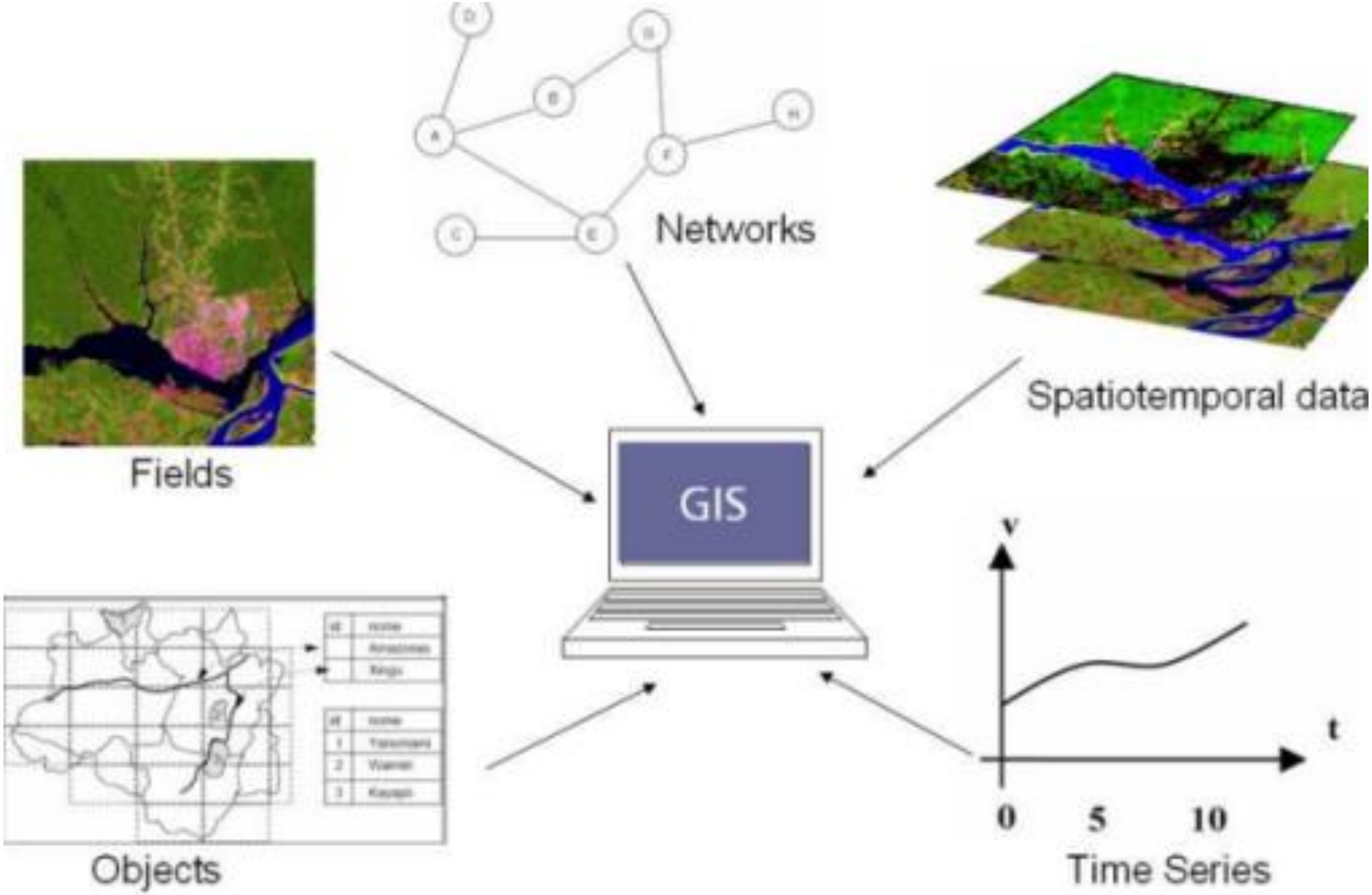

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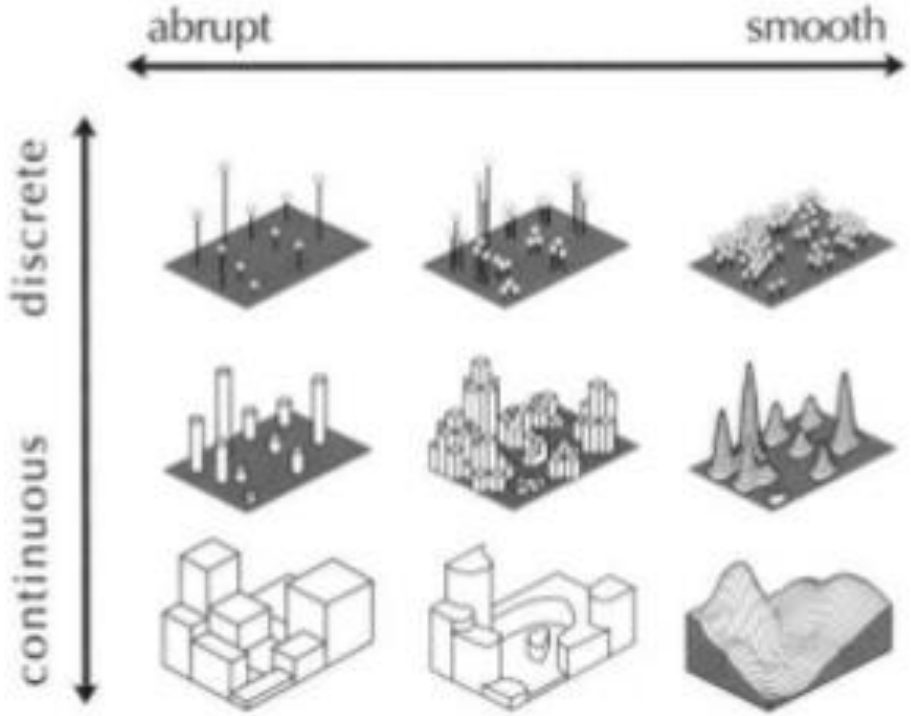
GEOGRAPHICAL PHENOMENON

- Events take place in geographical space and time
- Choice for a digital representation depends on the type of phenomenon
- Can be:
    - Artificial: Buildings
    - Natural: Rivers
    - Mixed Type: Pollution



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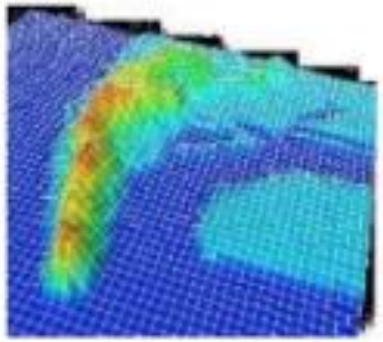
Groups of Geographical Phenomenon

Two common groups of geographic phenomena:

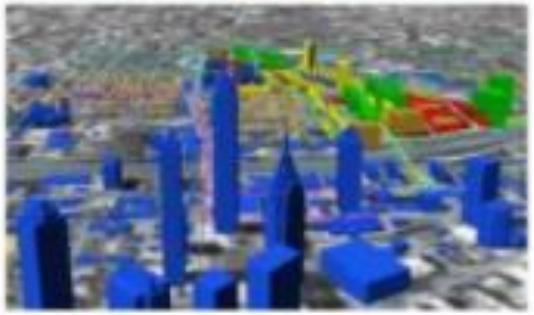
Fields

- For every point in the study area, a value can be determined.
- All changes in field values are gradual.
- Example: Elevation, Temperature.



Objects

- Well distinguishable discrete entities.
- Empty spaces in between the features.
- Study space grouped into mutually exclusive bounded parts.
- Example: Buildings, Roads



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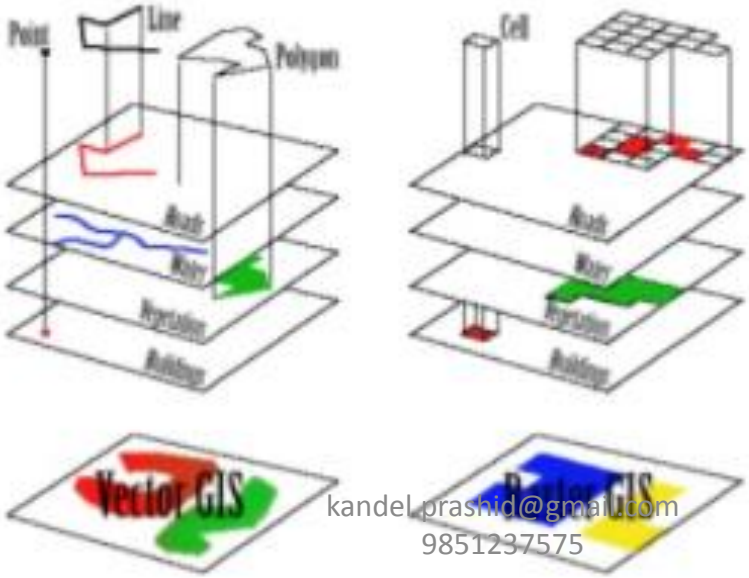
Putting your data into GIS

- ❑ What original raw data is available?
- ❑ What sort of data manipulation does the application want to perform?



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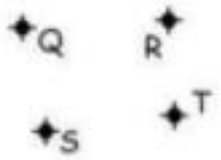
Data Representation on Computer

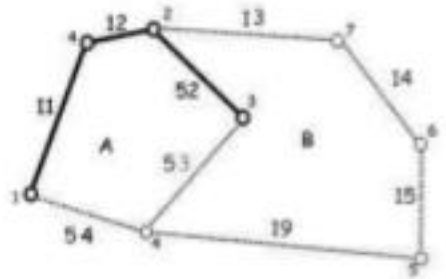
Vector Representations

- Represents each Geographic feature by a set of coordinates.
- An attempt to represent the object as exactly as possible.
- All positions, lengths, and dimensions to be defined precisely.

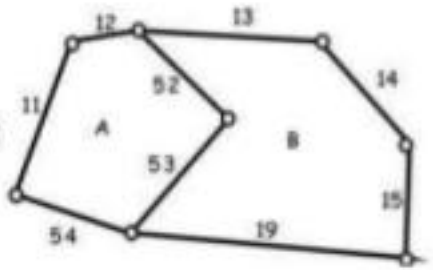
Points



Lines



Polygons



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Types:

- Point representation
  - Defined as single coordinate pairs (x,y) in 2D and (x, y, z) in 3D

- Line Representation
  - Defined by 2 end nodes and 0-n internal nodes in between

- Area Representation
  - Represent each polygon as a set of XY co-ordinates of the boundary
  -

- TIN Representation

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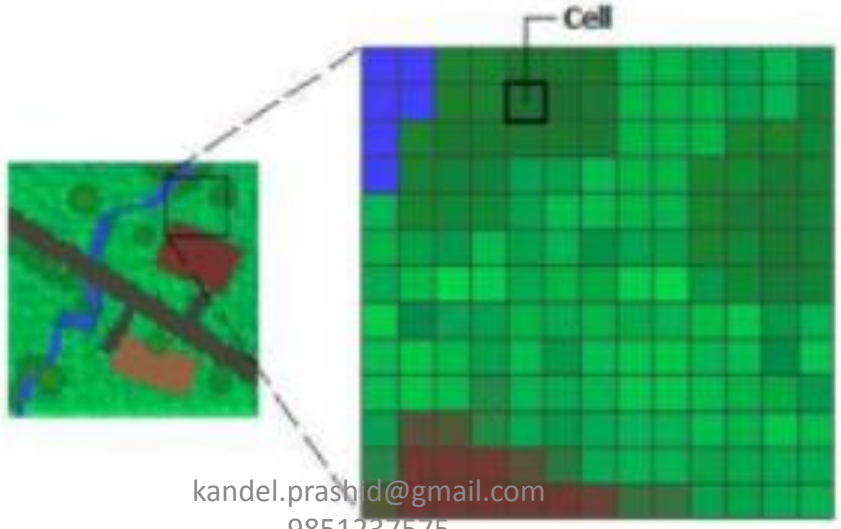
Raster Representation

- Entire space is broken into grid cells of a fixed or uniform size.
- Used Commonly to represent Geographic Fields.
- Each grid cell is referenced by a row and column number.



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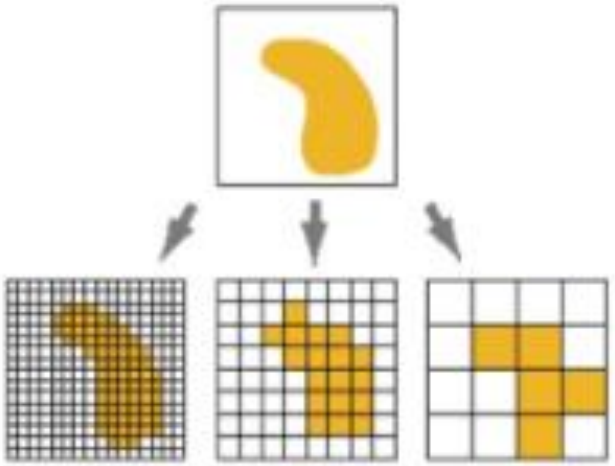
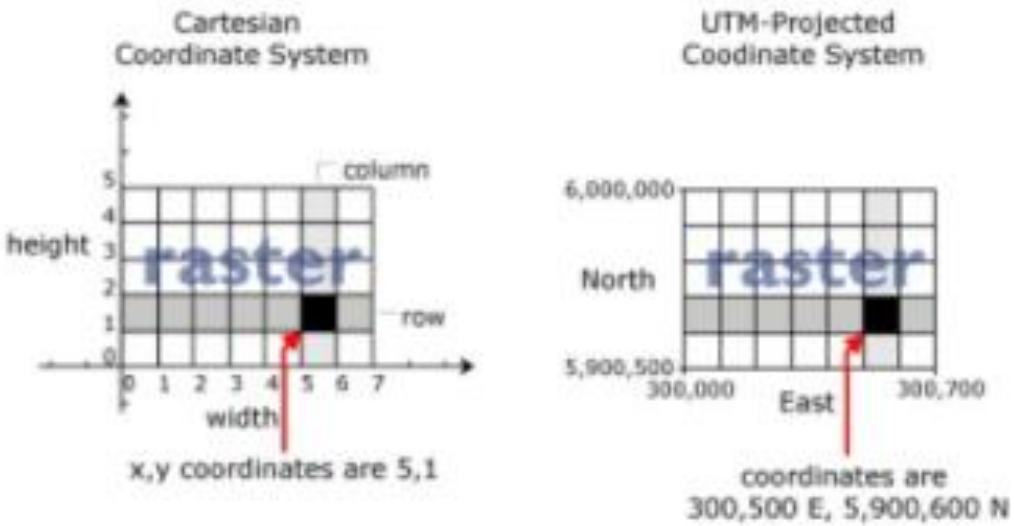
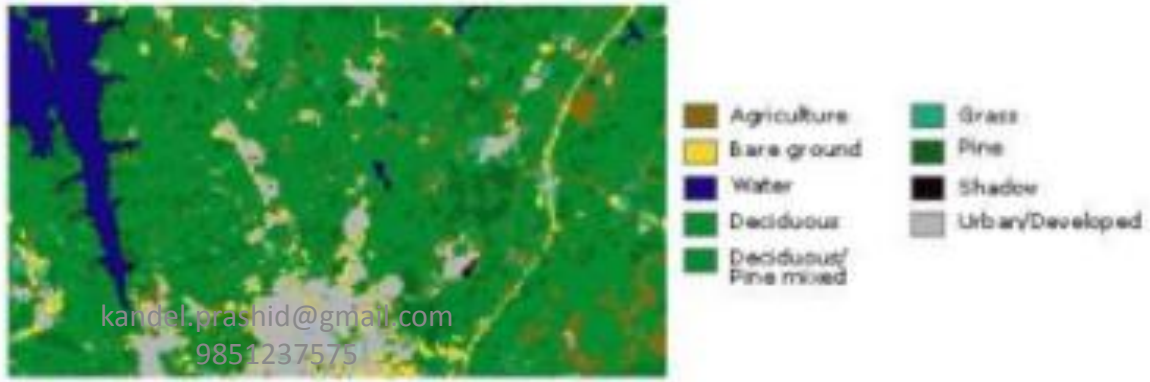


Raster representation with different Cell Size



Associating Coordinate System to a Raster data

Example:
Use of raster data as a thematic data, Land Use



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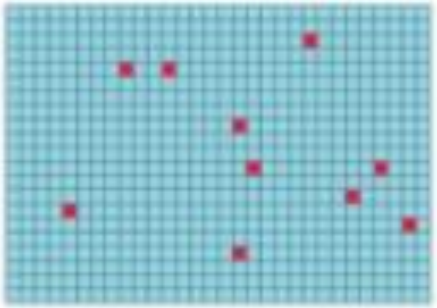
- A point is represented by a single grid cell.

Point features



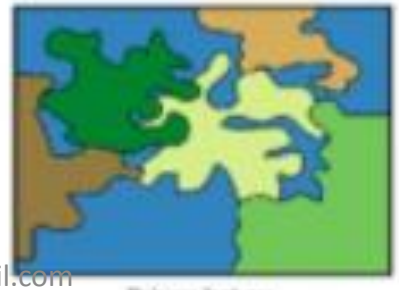
Raster point features

- A line by a number of neighboring cells strung out in given direction.

Line features

Raster line features

- An area by an agglomeration of neighboring cells.



Polygon features

Raster polygon features

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VECTOR	RASTER
❑Complex data Structure	❑Simple Data Structure
❑Easy association with attribute data	❑Difficult to associate attribute data
❑Efficient representation of topology	❑Difficult in representing topology.
❑Overlay of several vector polygon maps creates difficulties	❑Overlay and combination of data is easier

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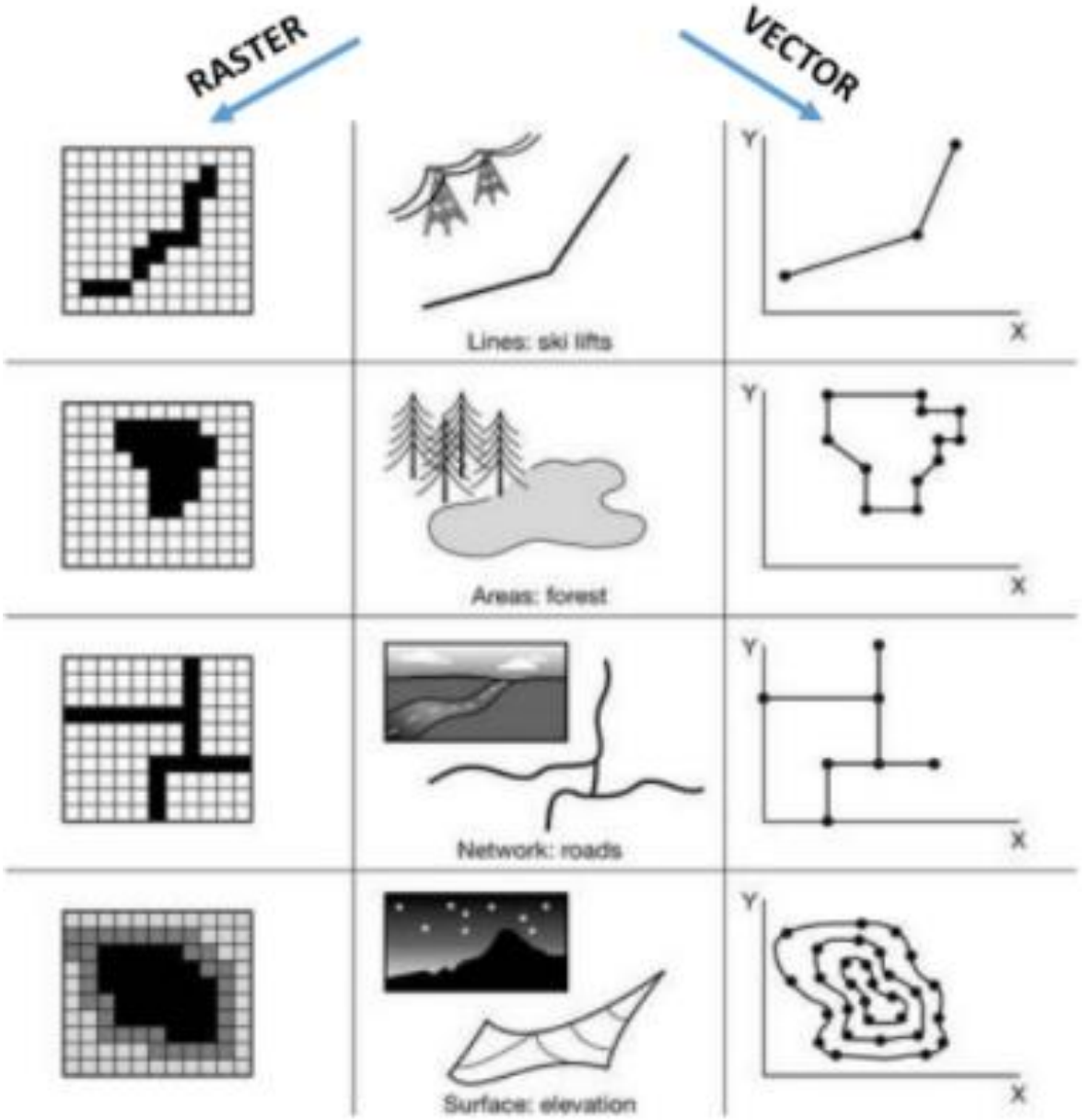
REAL WORLD



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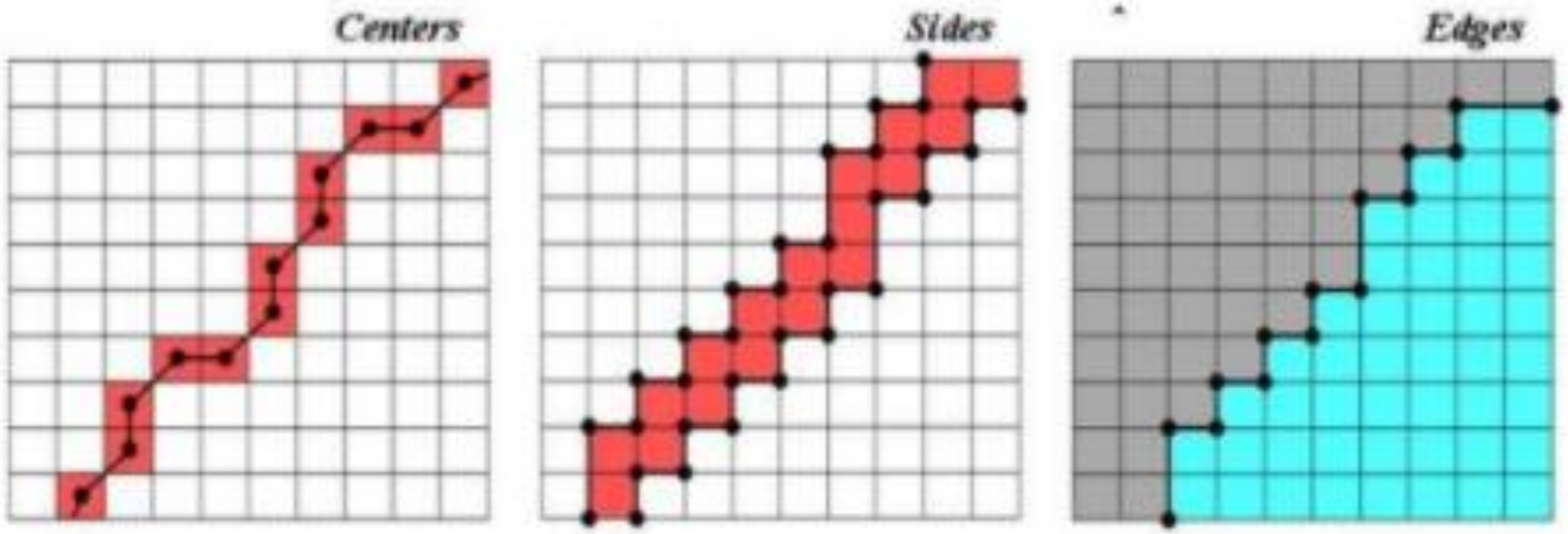
- Both structure are inter-convertible.
- Conversion from Vector to Raster
  - Is simplest
  - Many well known algorithms exist.
- Conversion from Raster to Vector
  - Much more complex operation.



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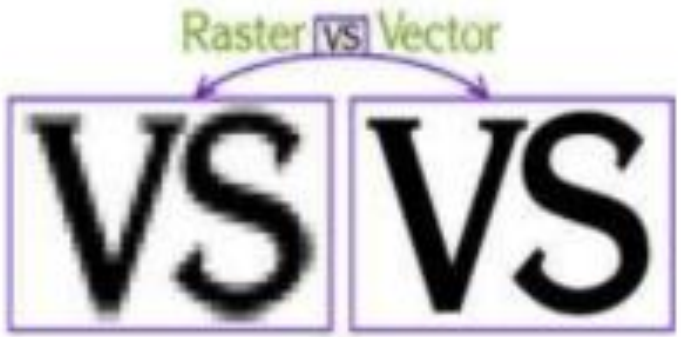
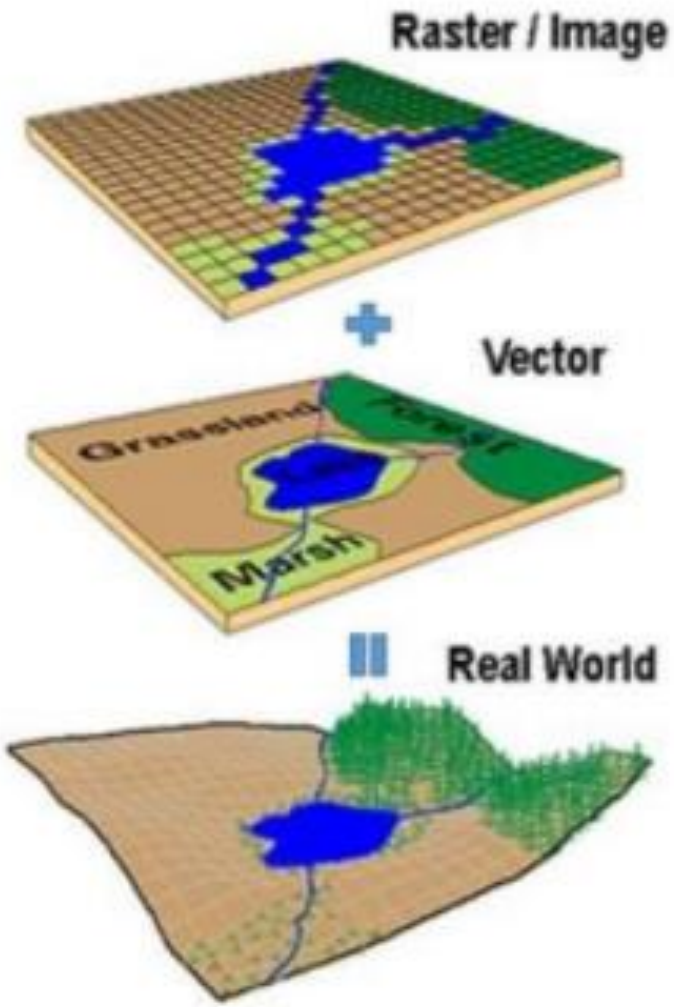
Choice between Vector and Raster

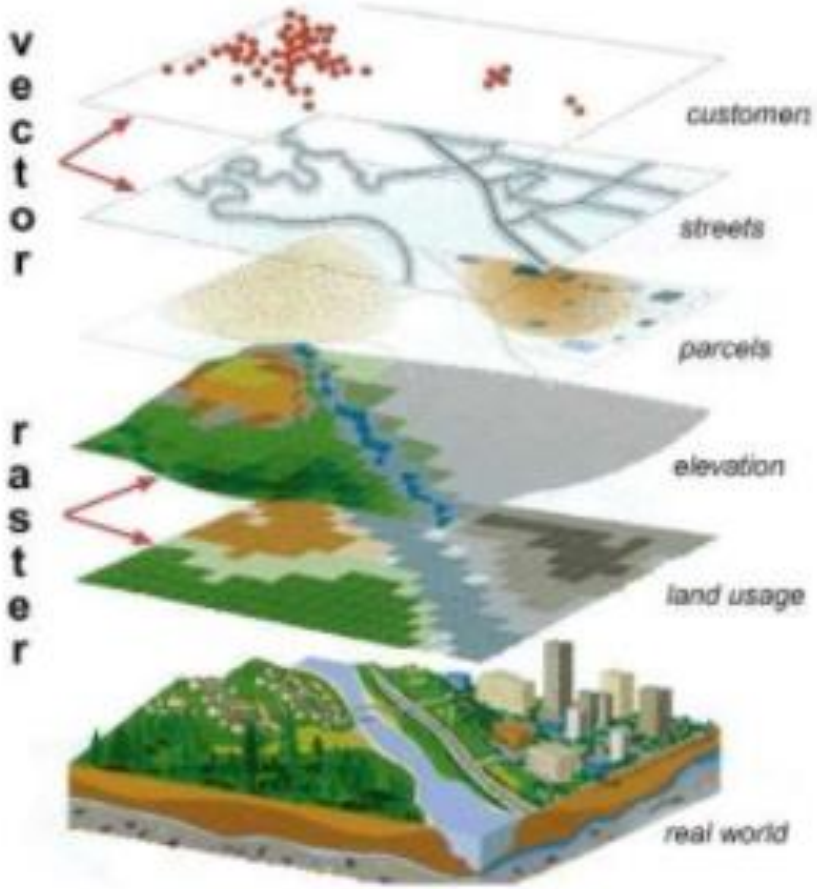


- Use VECTOR data structures for
  - Geographical objects like buildings ,Roads etc.
  - Network analysis :Transportation networks, Telephone networks etc.

- Use RASTER methods
  - Representing continuous geographical fields
  - When it is necessary to work with surfaces; Simulation and modeling
  - Spatial analysis ,Map overlays, etc.

Conclusion





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SPATIAL RELATIONSHIP AND TOPOLOGY

- In a GIS, spatial relationships are named "topology". Topological relations are made up of the geometric primitives: Point (simplest element), line (connected points), polygons (connected lines) *(**ESRI**)*. Based on these structures, the system is able to identify topological relationships and perform analyses.
- Topological arrangements of objects are accessed through features such as "**next**", "**part of**" or "**within**".
- Topology deals with spatial and structural properties of geometric objects, independent of their extension, type, or geometric form. Among the types of topological properties of objects there are: the number of dimensions an object has or the relationships that exist between objects.

- **VECTOR** An interesting method for the classification of topological relations was proposed by Egenhofer *(1993)(Worboys et al. 2004)*. It is called the 9-intersection schema.
- This intersection scheme is an elegant approach for the classification of topological configurations.
- The basic idea is based on the concept that each element is composed of a boundary (b), an interior (i), and an exterior (e). The concept of interior, boundary and complement (exterior) are defined in the general topology.

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Boundary

- The boundary consists of points or lines that separate the interior from the exterior. The edge of a line consists of the endpoints. The boundary of a polygon is the line that defines the perimeter.

Interior

- The interior of an object consists of points, lines or areas that are in the object but do not belong boundary.

Complement

- The complement, also called exterior, consists of the points, lines and areas which are not in the object.

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Types of Topology

Line features can share endpoints

arc-node topology

Area features can overlap with other area features

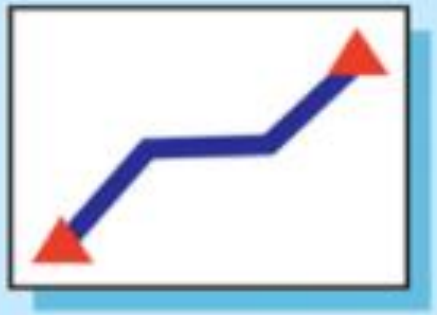
region topology

Area features can share boundaries

polygon topology

Line features can share endpoint vertices with point features

node topology



Line features can share segments with other line features

route topology

Point features can share vertices with line features

point events

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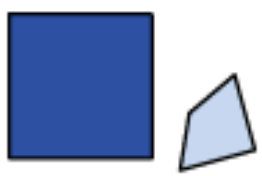
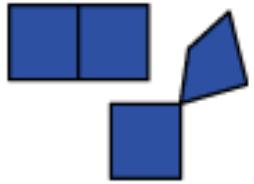
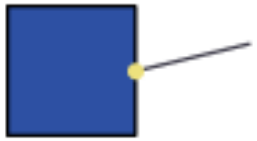
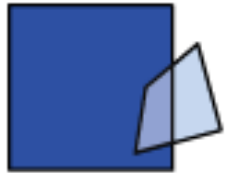
- **Disjoint**
- *There is no intersection area between object A and object B.* Test for disjoint.
- **Meet**
- *Object A and object B meet at the boundary.* The boundaries meet, but not the interior. Two geometry objects meet if the boundaries touch. Test for touch.
- **Overlap**
- *Object A and object B overlap*. Test for intersect (inversion of disjoint).
  Overlap with disjoint: The interior of an object intersects the boundary and the interior of the other object, but the boundaries do not intersect. That is the case if a line starts outside a polygon (area) and ends in the interior of the polygon.
  Overlap with Intersect: The boundaries and the interior of both objects intersect. If a geometry object has to intersect another geometry object the geometry needs to be part of the dimension of the bigger object. That means:

- *Points*
  - Cannot intersect with points, lines or areas.
- *Lines*
  - Cannot intersect with points.
  - Can intersect with other lines » intersection = point.
  - Can intersect with polygons » intersection = lines (or points).

- **Contains**
- *Object A contains object B.* Test whether the initial geometry object encloses a different geometry object. The interior and the boundary of an object are completely inside of the other object. A geometry object cannot contain a geometry object of higher order. E.g.:
- Points can not contain lines or polygons.
- Lines can not contain polygons.
- **Inside**
- *Object B lies inside object A.* It is the opposite of "contain". If A is inside B, then B contains A.

- **Covers**
- *Object A covers object B.* The interior of an object is completely inside the other object and the boundaries intersect. A geometry object can’t include a geometry object of higher dimension. That means:
- Points can not contain lines or polygons.
- Lines can not contain polygons.
- **Covered by**
- *Object B is covered by object A.* It is the opposite of "covers". If A is covered by B, then B covers A.
- **Equal**
- *Object B and object A match.* Test for equality of the initial geometry object and a different geometry object. The interior and the boundary of an object are lying on the boundary of the other object and vice versa. This happens when a line falls exactly on the boundary of a polygon. The coordinates of all components have to be equal. The compared geometry objects must be equal. That means:
- Point = point
- Line = line
- Polygon = polygon

The Most Common Topological Relationship

	poly-poly	line-line	point-point	poly-line	poly-point	line-point
Disjoint						
Meet						
Overlap						
Contains						

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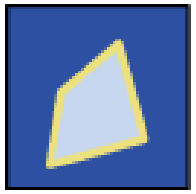
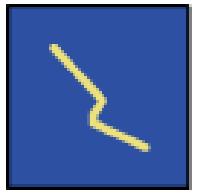
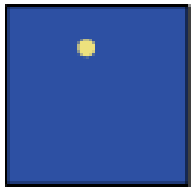
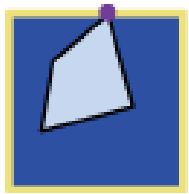
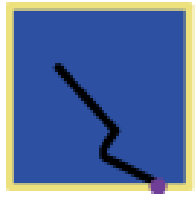
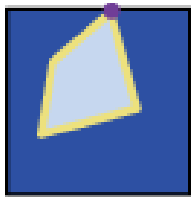
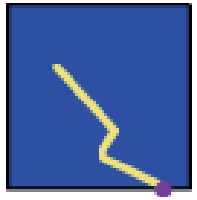
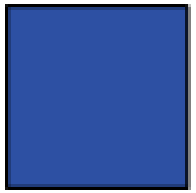
Inside						
Covers						
Covered by						
Equal						

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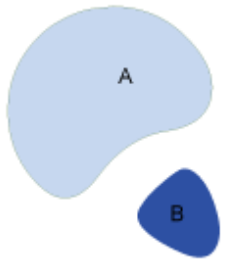
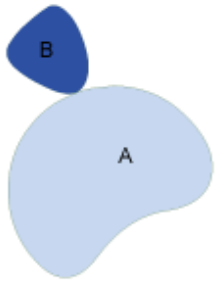
9-intersetion-schema and the 4-intersection-schema for some typical topological relations between two polygons, proposed by Egenhofer et al. (*1993*).

Topological relation	Graphical description	4-intersection-matrix	9-intersection-matrix
Disjoint		Objekt B Objekt A $\begin{array}{c\|cc} & b & i \\ \hline b & 0 & 0 \\ i & 0 & 0 \end{array}$	Objekt B Objekt A $\begin{array}{c\|ccc} & b & i & e \\ \hline b & 0 & 0 & 1 \\ i & 0 & 0 & 1 \\ e & 1 & 1 & 1 \end{array}$
Meet		Objekt B Objekt A $\begin{array}{c\|cc} & b & i \\ \hline b & 1 & 0 \\ i & 0 & 0 \end{array}$	Objekt B Objekt A $\begin{array}{c\|ccc} & b & i & e \\ \hline b & 1 & 0 & 1 \\ i & 0 & 0 & 1 \\ e & 1 & 1 & 1 \end{array}$
Overlap		Objekt B Objekt A $\begin{array}{c\|cc} & b & i \\ \hline b & 1 & 1 \\ i & 1 & 1 \end{array}$	Objekt B Objekt A $\begin{array}{c\|ccc} & b & i & e \\ \hline b & 1 & 1 & 1 \\ i & 1 & 1 & 1 \\ e & 1 & 1 & 1 \end{array}$

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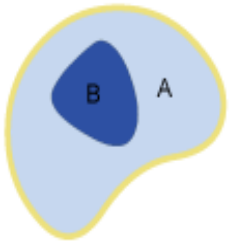
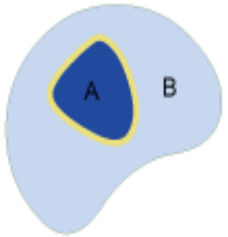
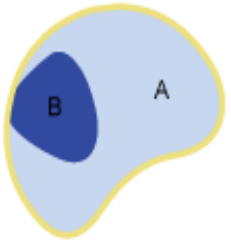
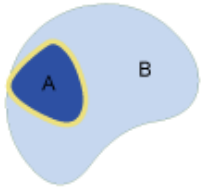
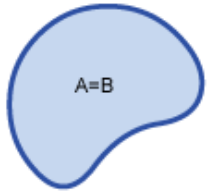
Contains		Objekt B b i Objekt A b [0 0] i [1 1]	Objekt B b i e Objekt A b [0 0 1] i [1 1 1] e [0 0 1]
Inside		Objekt B b i Objekt A b [0 1] i [0 1]	Objekt B b i e Objekt A b [0 1 0] i [0 1 0] e [1 1 1]
Covers		Objekt B b i Objekt A b [1 1] i [0 1]	Objekt B b i e Objekt A b [1 0 1] i [1 1 1] e [0 0 1]

Covered by		Objekt B b i Objekt A b [1 1] i [0 1]	Objekt B b i e Objekt A b [1 0 1] i [1 1 1] e [0 0 1]
Equal		Objekt B b i Objekt A b [1 0] i [0 1]	Objekt B b i e Objekt A b [1 0 0] i [0 1 0] e [0 0 1]

http://www.gitta.info/SpatialQueries/en/html/TopoBasedOps_learningObject1.html

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Scale and Resolution

See it on Unit 2

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UNIT 4 – CAPTURING THE REAL WORLD

- Different methods of data capture
- Map projection and spatial reference
- Data preparation, conversion and integration
- Quality aspects of spatial data
- GPS
- Remote Sensing

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GIS Data Creation/Data Capture

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GIS Data Creation

- GIS data creation is the extraction in digital form of real world data to be used for analysis or visualization. It helps explain the existing scenarios or model in reference to the geographical location of targets.
- GIS data is the basis for such decision making on a GIS platform.
- The power of a GIS is in the data analysis. Without data, all the bells and whistles of GIS are just that.
- The effort to bring stable, accurate data is an enormous one for any GIS. Data development and maintenance is the most costly, labor intensive part of developing a GIS.
- There are several ways in which to bring spatial data into a GIS.
- Quality data is essential to conduct robust analysis and to produce a meaningful map.

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- There are two major types of GIS data; “Vector Data” and “Raster Data”.
- Vector data uses with geometric baseline to represent the data from the perspectives of the cartographers.
- Raster data on the other had is cell-based data that are stored in picture format using digital number value for each cell of the image.
- In order to produce GIS data, valid spatial and attribute data is essential, and this involves a software input system such as ArcGIS, QGIS, ERDAS, as well as potentially hardware supports such as GPS devices.
- GIS data creation is a multi stage process (i.e. Geo-referencing, Digitizing, GPS data integration, or coordinate data import, attribute table import, and join data with spatial extension) and it requires careful management and nurturing to create a reliable and accurate database.

GIS data Creation: Manual Digitizing

- This is traditionally the most common way to convert paper-based sources of spatial information (e.g. maps) to digital data. The paper map is attached by tape to a digitizing table (or tablet as the smaller digitizers are known).
- The data is then digitized by tracing the features of interest with a mouse like hand held device called a puck.

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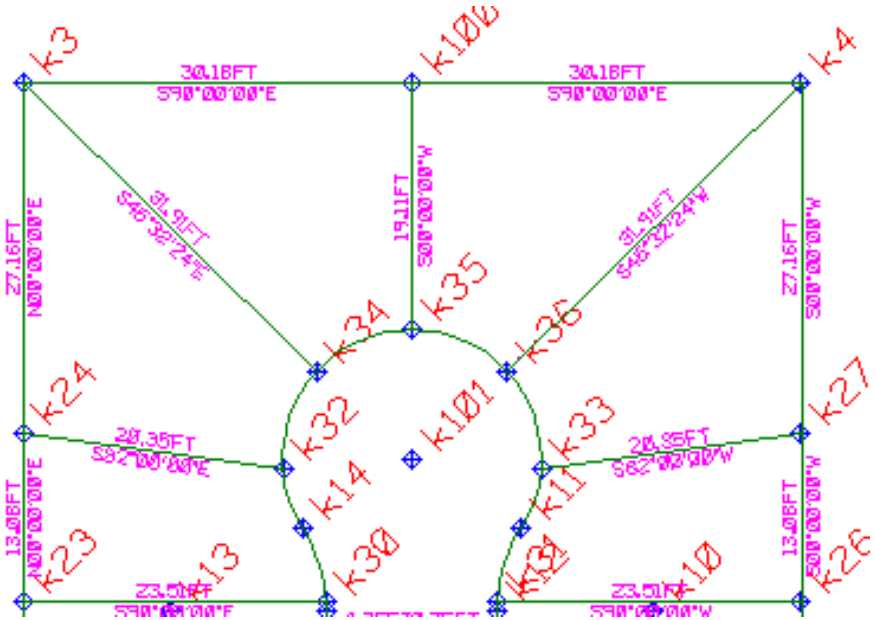
- The data is then digitized by tracing the features of interest with a mouse like hand held device called a puck.

Heads up Digitizing

- With the proliferation of low cost sources of digital imagery and large format scanners, heads up digitizing is becoming a popular method of digital conversion.
- Also know as on-screen digitizing, this method involves digitizing directly on top of an orthorectified image such as a satellite image or an aerial photograph.
- The features of interest are traced from the image. The benefit of this over manual digitizing is that no transformation is needed to convert the data into the needed projection.
- In addition, the level of accuracy of the derived dataset is taken from the initial accuracy of the digital image.
- Heads-up digitizing is also utilized in extracting data from scanned and referenced maps.

Coordinate Geometry (COGO)

- Coordinate geometry is a keyboard-based method of spatial data entry.
- This method is most commonly used to enter cadastral or land record data.
- This method is highly precise as entering the actual survey measurements of the property lines creates the database.
- Distances and bearings are entered into the GIS from the original surveyor plats. The GIS software builds the digital vector file from these values.



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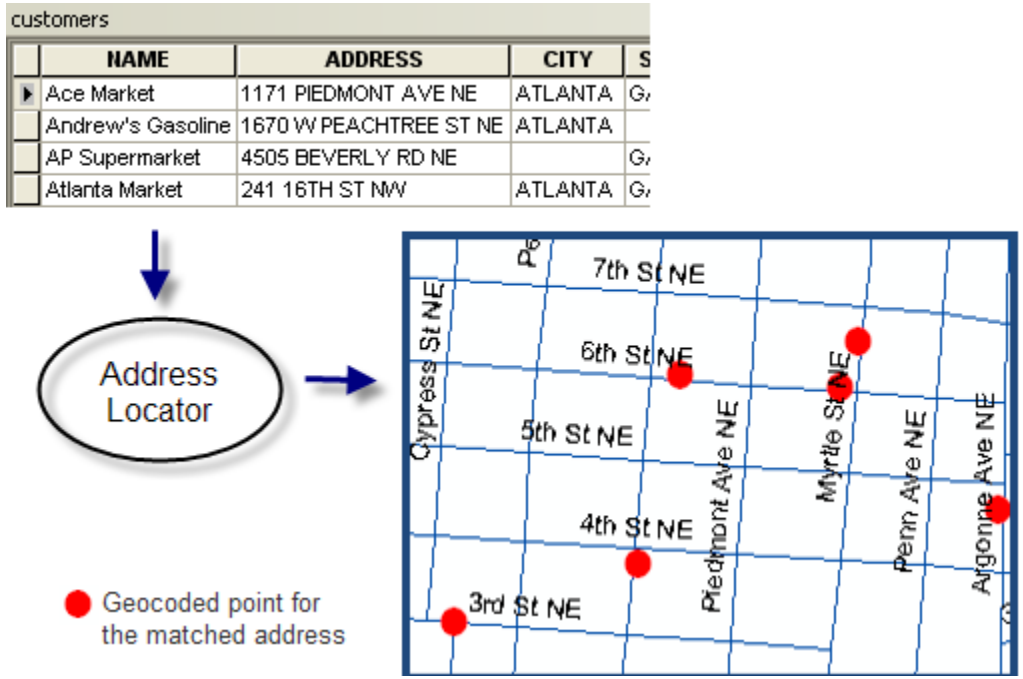
Geocoding

- Geocoding is another keyboard-based method. Geocoding uses addresses from a flat file (such as a .dbf file, MS Access database or excel spreadsheet) to create x,y coordinate locations interpolated from a geocodable spatial database.
- These spatial databases are most commonly street centerline files.
- The resultant geocoded database is a point file.



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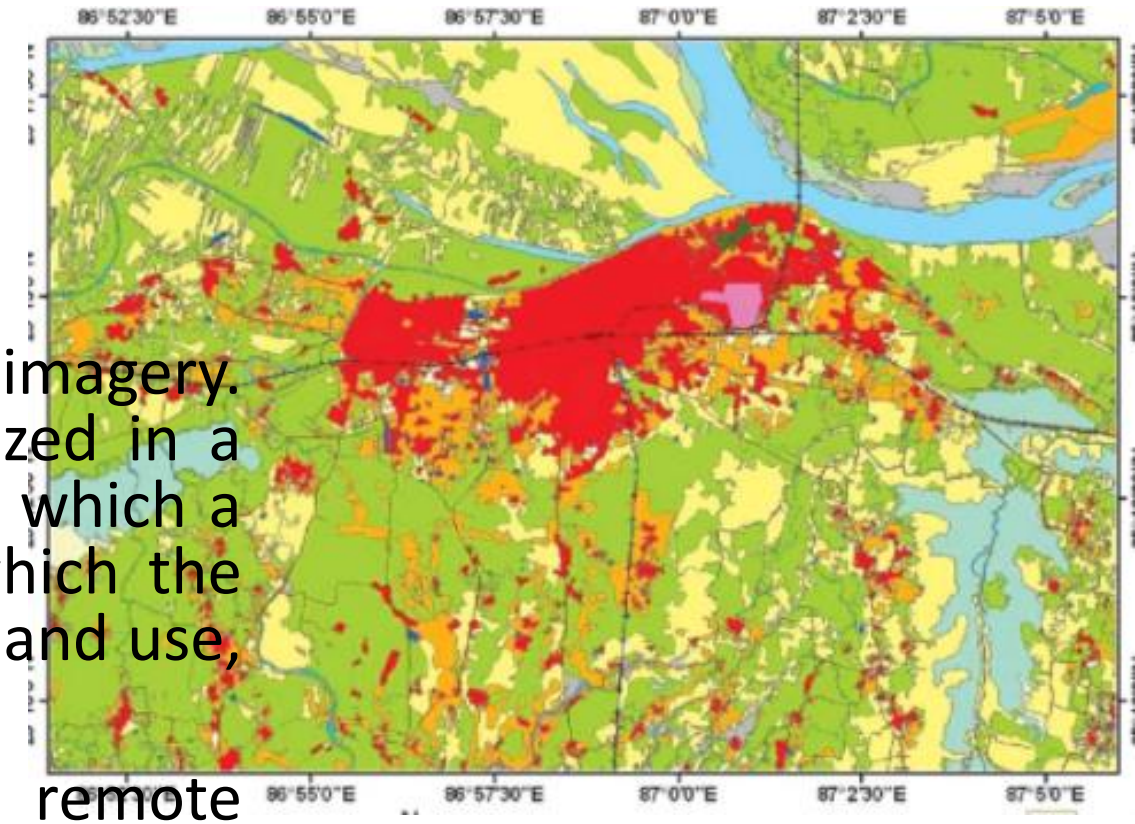
GPS

- GPS is a way to gather accurate linear and point location data. Originally devised in the 1970s by the Department of Defense for military purposes, the current GPS consists of 28 satellites that orbit the earth, transmitting navigational signals.
- Through interpolation, these signals received by a data logger can pinpoint the holder's location.
- Depending on the unit, the locational accuracy can reach to the millimeter.
- Combined with attribute data entered at the time of collection, GPS is a rapid and acccurate method of data collection.

Image Processing



- Geodatasets can be derived from digital imagery. Most commonly satellite imagery is utilized in a process called supervised classification in which a user selected a sampling of pixels for which the user knows the type (vegetation species, land use, etc).
- Using a classification algorithm, remote sensing software such as ERDAS or ENVI classifies a digital image into these named categories based on the sample pixels. In contrast to the other methods discussed, supervised classification results in a raster dataset.

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Precaution in Data Capture:

There are many tools that incorporate manual skills to capture the data using various softwares like ArcGIS. These softwares allows user to capture information from the base data.

- During this data capture, the user may misinterpret features from the base data and captures the features with errors. For example: A user misinterprets two buildings as single building and capture as a single feature. But in real world, there are two features.
- So, the correct interpretation of features in base data must be performed.
- However, there are many tools that enables user to find and fix those errors, but still these tools are not used frequently due to lack of awareness.
- Data capture must be performed on a perfect scale where one must be able to view the features distinctly.

Spatial Reference System and Projection

- A **spatial reference system** (**SRS**) or **coordinate reference system** (**CRS**) is a coordinate-based local, regional or global system used to locate geographical entities.
- A spatial reference system defines a specific map projection, as well as transformations between different spatial reference systems.
- Spatial reference systems are defined by the OGC's (Open Geospatial Consortium) Simple feature access using well-known text, and support has been implemented by several standards-based geographic information systems.
- A spatial reference is a series of parameters that define the coordinate system and other spatial properties for each dataset in the geodatabase. It is typical that all datasets for the same area (and in the same geodatabase) use a common spatial reference definition.

- A spatial reference includes the following:
  - The coordinate system
  - The coordinate precision with which coordinates are stored (often referred to as the coordinate resolution)
  - Processing tolerances (such as the cluster tolerance)
  - The spatial extent covered by the dataset (often referred to as the spatial domain)

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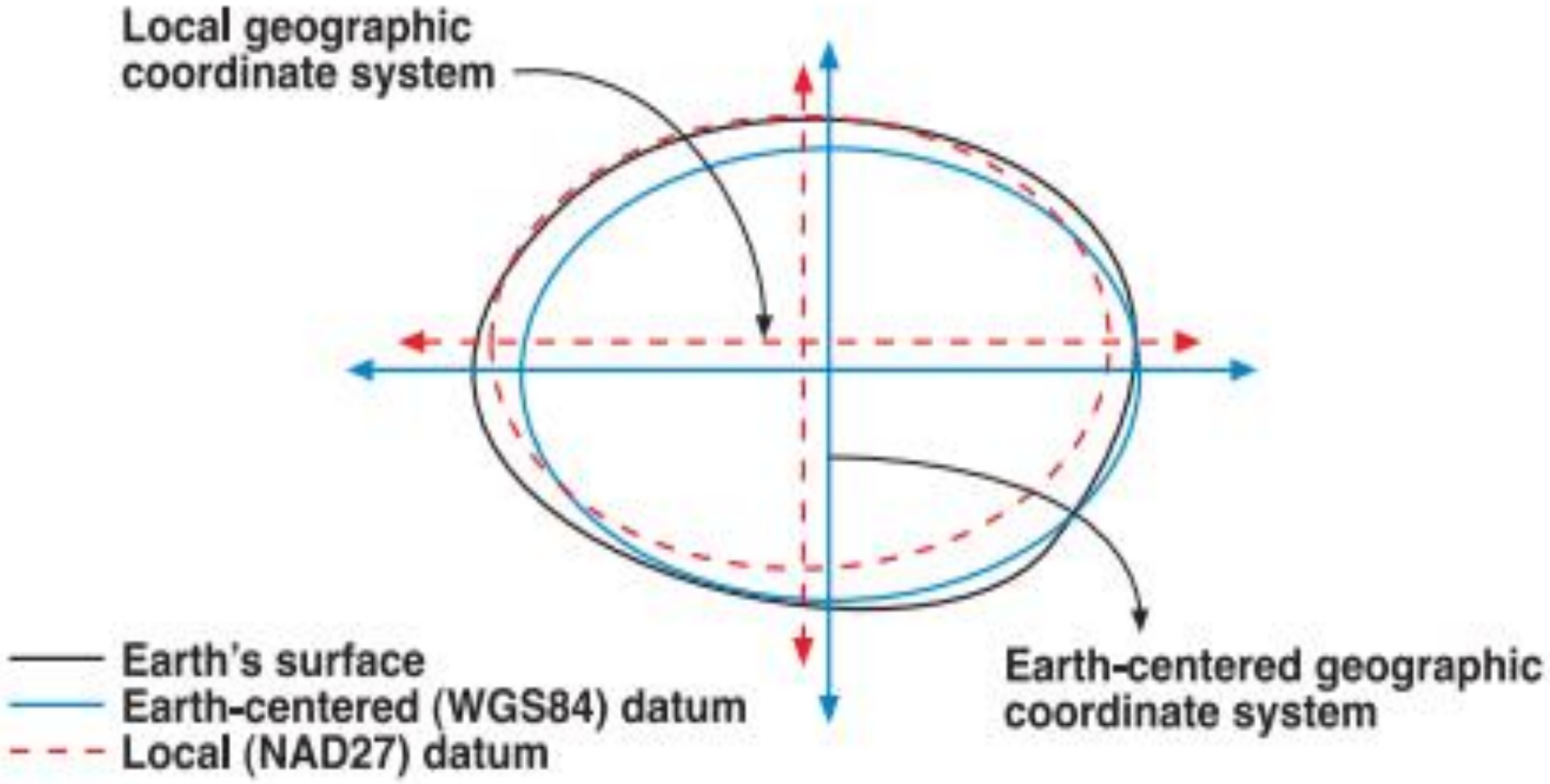
MAP Projection

- Is very complex topic in GIS
- A map projection is a mathematical expression that is used to represent the round and3D surface of the earth on a flat and 2D map.

- A projected coordinate system, unlike a geographic one, has the advantage that lengths, angles, and areas are constant across the two dimensions. This is not true when working in a geographic coordinate system. A projected coordinate system is always based on a geographic coordinate system that can use a sphere or spheroid.


Local geographic coordinate system
Earth's surface
Earth-centered (WGS84) datum
Local (NAD27) datum
Earth-centered geographic coordinate system

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- A spheroid can’t be flattened to a plane any easier than flattening a piece of orange peel—it will rip.
- Similarly, representing the earth’s surface in two dimensions causes distortion in the shape, area, distance, or direction of the data.

- Different projections cause different types of distortions. Some projections are designed to minimize the distortion of one or two of the data’s characteristics. A projection could maintain the area of a feature but alter its shape.
- **Map projections** portray the surface of the earth on a two-dimensional, flat piece of paper or computer screen.

Real World Coordinate System

- A coordinate system is a system for assigning tuple (order list) for a numbers of each point in an n-dimensional space. With the help of coordinate reference systems (CRS) every place on the earth can be specified by a set of three numbers, called coordinates.

- In general CRS can be divided into **projected coordinate reference systems** (also called Cartesian or rectangular coordinate reference systems) and **geographic coordinate reference systems** (also called Curivlinear.

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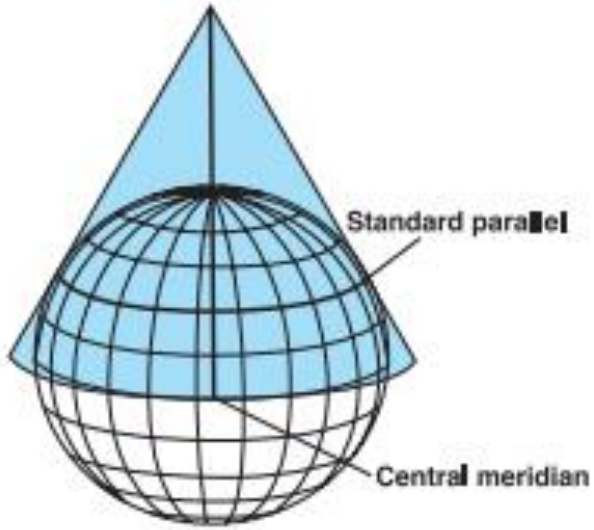
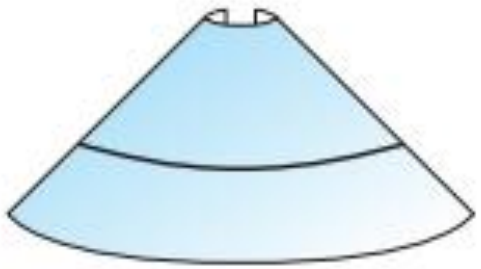
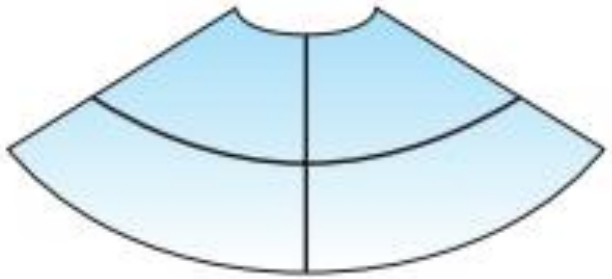
Projection types

- Many common map projections are classified according to the projection surface or geometric shapes used: conic, cylindrical, and planar.

1. **Conical Projection-**







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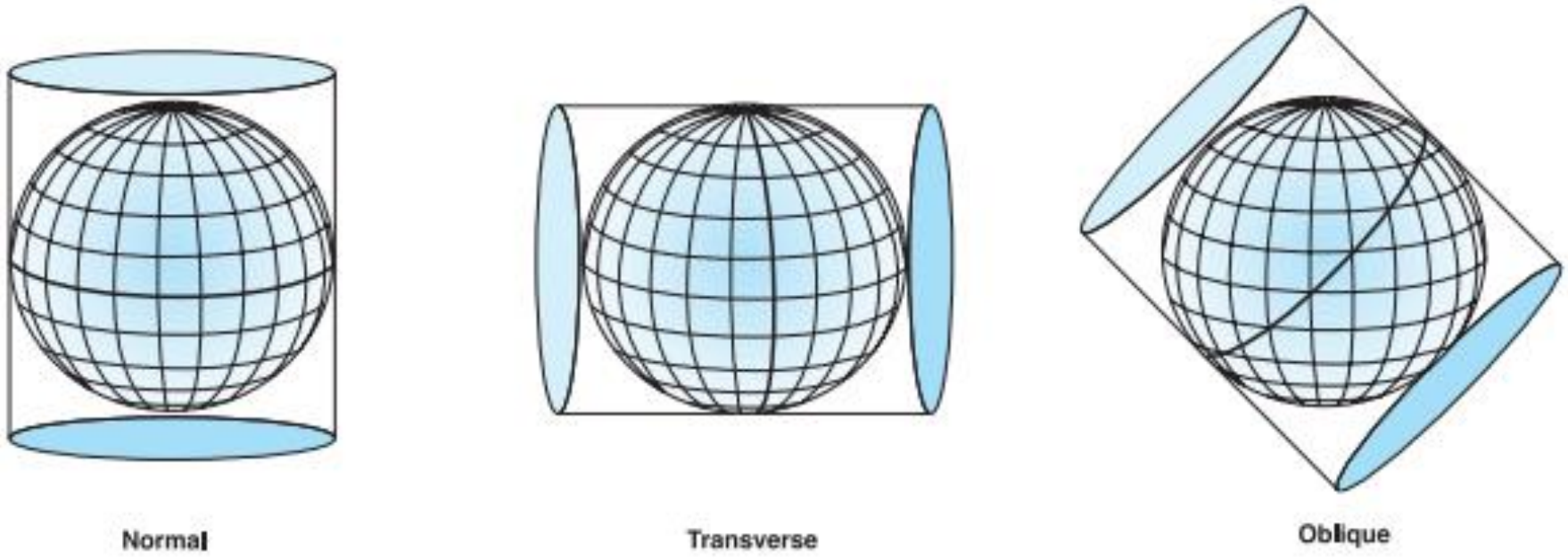
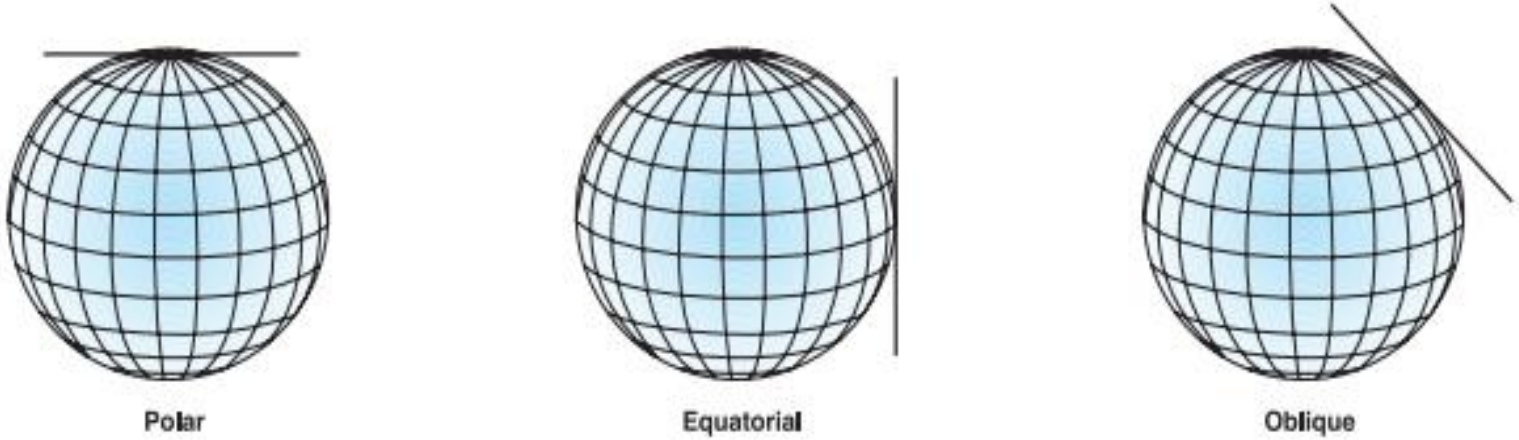
2. Cylindrical Projection



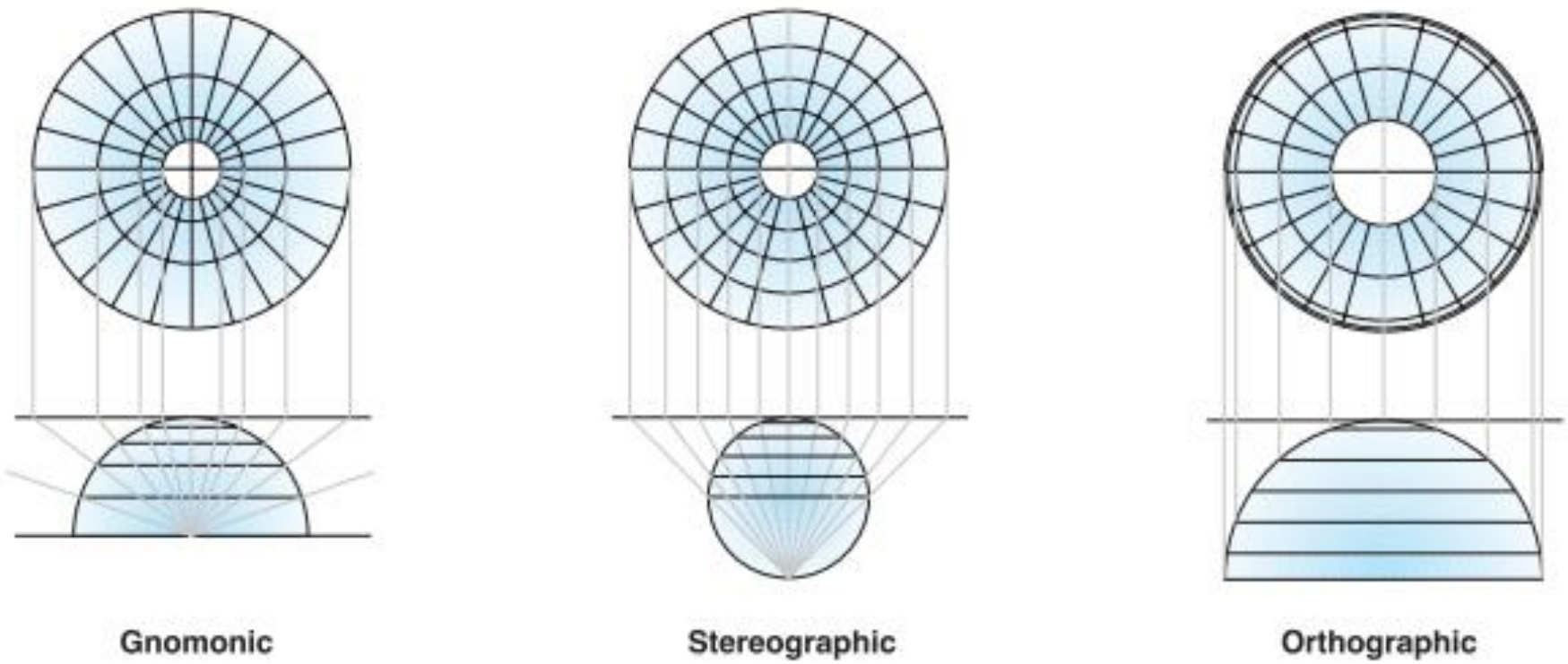
The Mercator projection is one of the most common cylindrical projections.

3. Planar Projection

- Planar projections project map data onto a flat surface touching the globe. A planar projection is also known as an azimuthal projection or a zenithal projection



- Some planar projections view surface data from a specific point in space. The point of view determines how the spherical data is projected onto the flat surface.



UTM Projection

- The UTM CRS is a global map projection. This means, it is generally used all over the world.
- To avoid too much distortion, the world is divided into **60 equal zones** that are all **6 degrees** wide in longitude from East to West.
- Parameters used in this projection are:-
  - False easting – 500000
  - False northing – 0
  - Scale Factor – 0.9996

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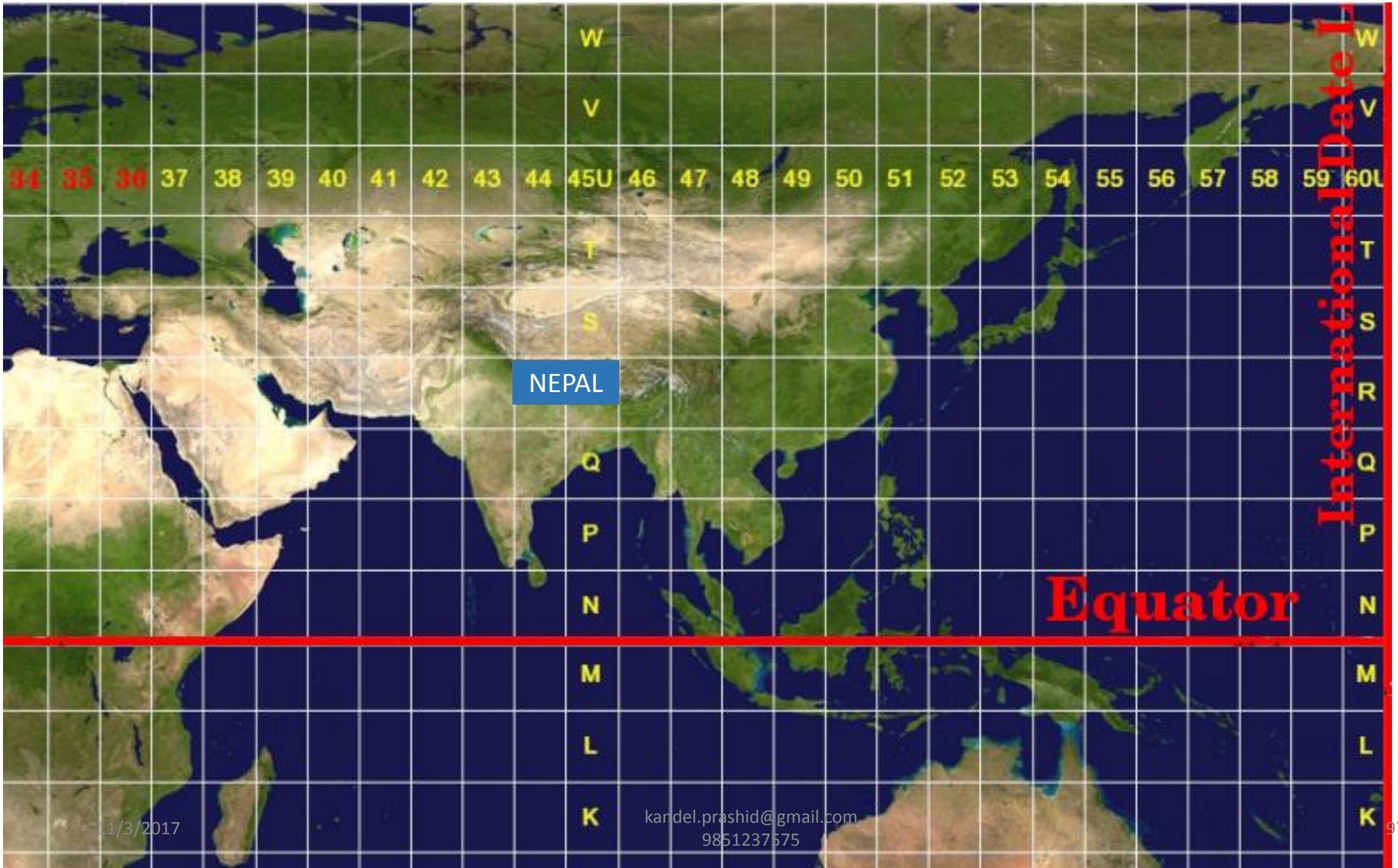

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Projection System Used in Nepal

- In recent years because Nepal has greater East-West expansion so as to increase accuracy for cadastral survey and various other purpose, UTM has been modified to **MUTM**(Modified UTM);Gauss-Krueger projection-based coordinate system, thus the “modified” designation.

- In this projection system Earth is divided in 120 zones each of 30.Nepal has central meridian of 810,840 and 870.The scale factor of 0.9999.
- Reference Spheroid – Everest 1830

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- Map projections are never absolutely accurate representations of the spherical earth.
- They show **distortions of angular conformity, distance and area.**
- It is impossible to preserve all these characteristics at the same time in a map projection.

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Data preparation

- In GIS, data preparation is very important for further analysis. While preparing the data we can choose various tool that is available in each GIS Software. For example:- the tools are available in ArcToolbox in ArcGIS of the ESRI.

- Data preparation involves important part of the GIS project. Some of the major data preparation are Georferences, Geocoding, Orthorectification, Registration, Projection, Transformation, Digitizing, Extract, Mask, Buffer, and Reclassification.
- These tools will assist in preparing the data according to our needs and requirement in the area of interest (AOI).

Data Conversion

- GIS data conversion transforms data from its current format (usually paper maps or related sources) into a digital format.
- In some cases, the data is already in a digital format, but often this data is not compatible with the current system.
- Conversion of data from one digital format to a new digital format is referred to as data translation. For example, digitizing a paper map into an electronic series of x, y coordinates is data conversion from paper to digital format.
- Similarly, In GIS another most important conversion is from raster to vector and vector to raster conversion.

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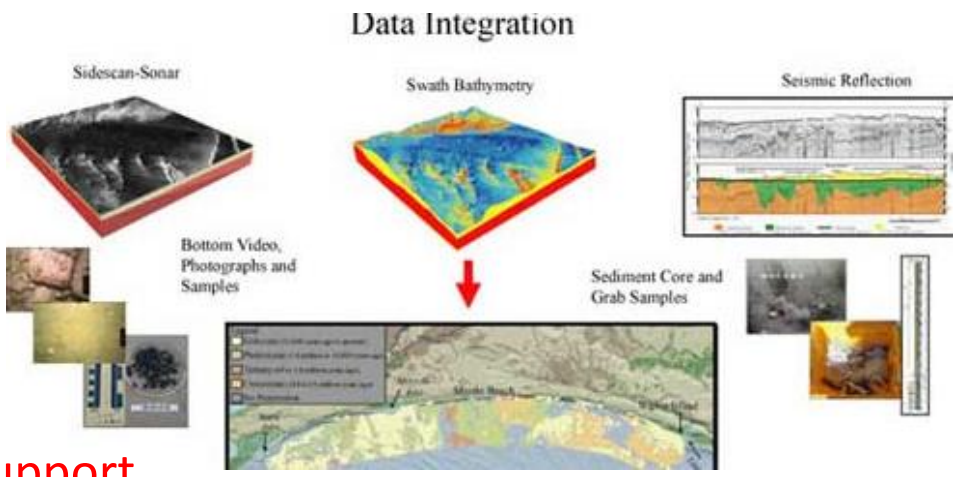
Data Integration

- **Data integration** involves combining data residing in different sources and providing users with a unified view of these data.
- **Data integration** fundamentally involves the combining or merging of data from multiple sources in an effort to extract better and/or more information. This may include data that are multitemporal, multiresolution, multisensor, or multi-data type in nature
- It is for the strategic planning and quick decision making.



Site Selection
Suitability Assessment
Favorability Assessment
Probability Assessment

Spatial Decision Support System
Multiple Map Modeling

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Quality aspects of spatial data

- Spatial data quality for many years has been considered important, it has not always been a priority. However, times are changing.
- In our day to day lives we are more and more depending on applications and services in which data plays a crucial role.
- And as more data becomes available we also have something to choose from. But how do we make that choice? How do we make sure that we end up using the best possible data set for our application?
- It is our view that spatial data quality has the potential to help us choose the correct dataset. Specifically the fitness for use approach can contribute in facilitating the choice in what data to use for a specific application.
- Data quality concepts provide an important framework for both data producers and users.

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- “The purpose of describing the quality of geographic data is to facilitate the comparison and selection of the dataset best suited to application needs or requirements’ (ISO 19157:2013).
- There are many standards describing quality, but since we are converging towards the ISO standards for spatial data we focus on the ISO 19157 standard describing spatial data quality by the following six groups of elements:
  - Completeness
  - Logical consistency
  - Positional accuracy
  - Thematic accuracy
  - Temporal quality and
  - Usability element

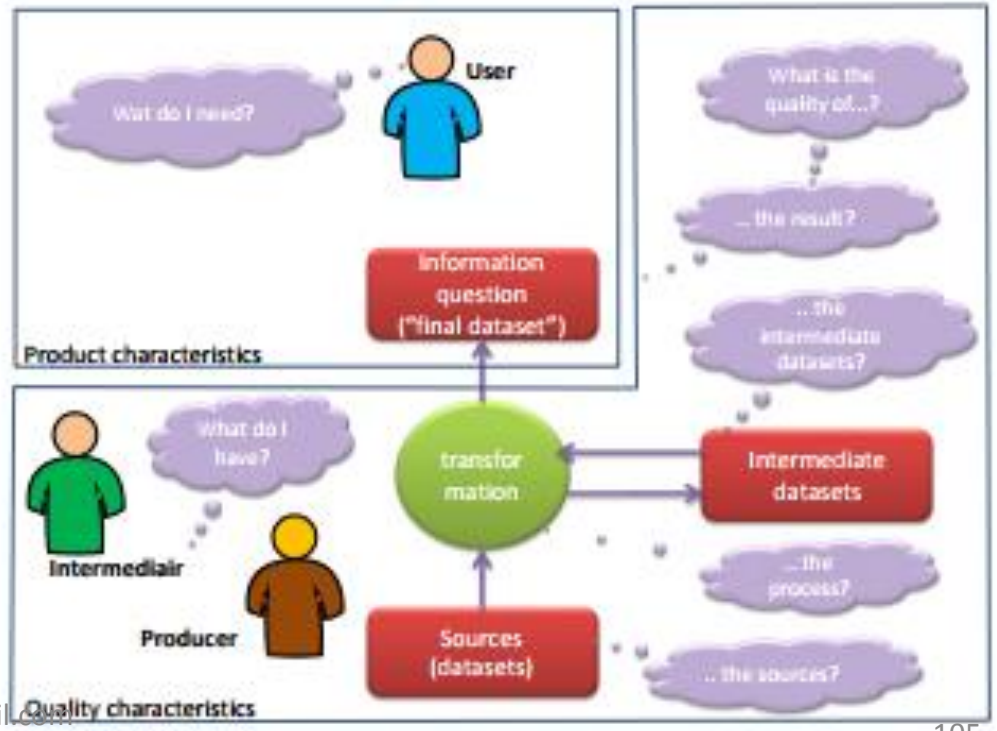

Figure 1: framework spatial data quality

- Lineage - Describes the history of the spatial data, including descriptions of the source material from which the data were derived, and the methods of derivation. It also contains the dates of the source material, and all transformations involved in producing the final digital files or map products.
- Positional accuracy – Refers to the absolute and relative accuracy of the positions of geographic features. Descriptions of positional accuracy include the quality of the final file or product after all transformations.
- Attribute accuracy - Refers to the accuracy of the quantitative and qualitative information attached to each feature.
- Logical consistency - Describes the dependability of relationships encoded in the data structure of the digital spatial data.
- Completeness - Refers to the degree to which geographic features, their attributes and their relationships are included or omitted in a dataset. It also includes information on selection criteria, definitions used, and other relevant mapping rules

GPS

35.9
0.1
Speed Limit 0.2
Downtown
Griffin St
65
60
9:00A
GARMIN
72
56
Last Sport
3.65
Time 23:25
Pace 6:25
7526

What is GPS ???

- The Global Positioning System (GPS) is a satellite-based navigation system made up of at least 24 satellites.
- GPS works in any weather conditions, anywhere in the world, 24 hours a day, with no subscription fees or setup charges.
- The U.S. Department of Defense (USDOD) originally put the satellites into orbit for military use, but they were made available for civilian use in the 1980s.
-

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How GPS Works ??

- GPS satellites circle the Earth twice a day in a precise orbit. Each satellite transmits a unique signal and orbital parameters that allow GPS devices to decode and compute the precise location of the satellite.
- GPS receivers use this information and trilateration to calculate a user's exact location.
- Essentially, the GPS receiver measures the distance to each satellite by the amount of time it takes to receive a transmitted signal. With distance measurements from a few more satellites, the receiver can determine a user's position and display it electronically to measure your running route, map a golf course, find a way home or adventure anywhere.

- To calculate your 2-D position (latitude and longitude) and track movement, a GPS receiver must be locked on to the signal of at least 3 satellites.
- With 4 or more satellites in view, the receiver can determine your 3-D position (latitude, longitude and altitude).
- Generally, a GPS receiver will track 8 or more satellites, but that depends on the time of day and where you are on the earth. Some devices can do all of that from your wrist.
- Once your position has been determined, the GPS unit can calculate other information, such as:

1. Speed 2. Bearing 3. Track 4. Trip distance 5. Distance to destination 6. Sunrise and sunset time 7. And more

Further Reading: *http://www8.garmin.com/aboutGPS/*

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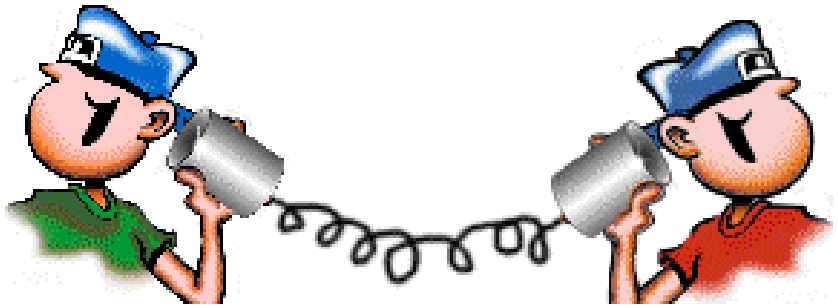
Remote Sensing



- *"Remote sensing is the science (and to some extent, art) of acquiring information about the Earth's surface without actually being in contact with it. This is done by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information."*
- In much of remote sensing, **the process** involves an interaction between incident radiation and the targets of interest.
- This is exemplified by the use of imaging systems where the following seven elements are involved.
- Note, however that remote sensing also involves the sensing of emitted energy and the use of non-imaging sensors.

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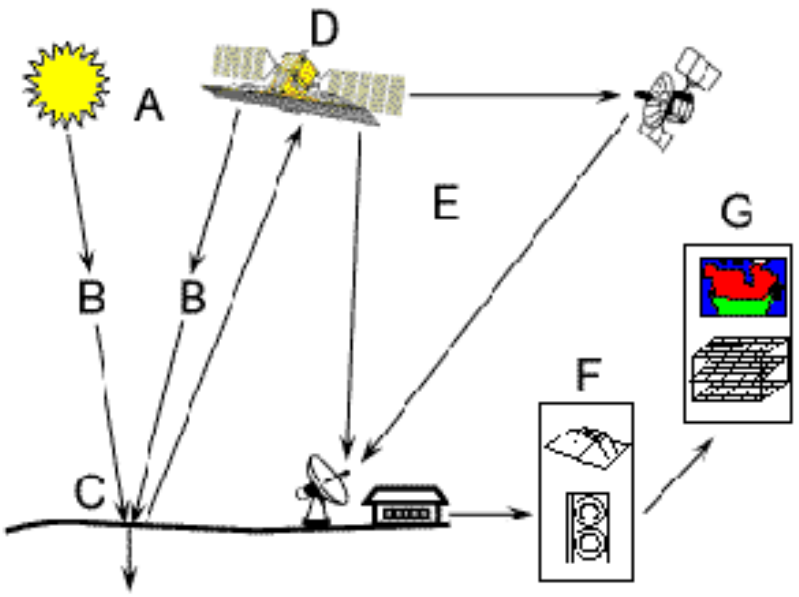
- **1. Energy Source or Illumination (A)** - the first requirement for remote sensing is to have an energy source which illuminates or provides electromagnetic energy to the target of interest.
- **2. Radiation and the Atmosphere (B)** - as the energy travels from its source to the target, it will come in contact with and interact with the atmosphere it passes through. This interaction may take place a second time as the energy travels from the target to the sensor.
- **3. Interaction with the Target (C)** - once the energy makes its way to the target through the atmosphere, it interacts with the target depending on the properties of both the target and the radiation.
- **4. Recording of Energy by the Sensor (D)** - after the energy has been scattered by, or emitted from the target, we require a sensor (remote - not in contact with the target) to collect and record the electromagnetic radiation.
- **5. Transmission, Reception, and Processing (E)** - the energy recorded by the sensor has to be transmitted, often in electronic form, to a receiving and processing station where the data are processed into an image (hardcopy and/or digital).
- **6. Interpretation and Analysis (F)** - the processed image is interpreted, visually and/or digitally or electronically, to extract information about the target which was illuminated.
- **7. Application (G)** - the final element of the remote sensing process is achieved when we apply the information we have been able to extract from the imagery about the target in order to better understand it, reveal some new information, or assist in solving a particular problem.
- These seven elements comprise the remote sensing process from beginning to end.



UNIT 5 – SPATIAL ANALYSIS AND VISUALIZATION

- Spatial Analysis :Overlay - Buffering
- Map outputs and its basic elements

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- Spatial analysis is the vital part of GIS. It can be done in two ways. One is the vector-based and the other is raster-based analysis.
- Two fundamental functions of GIS have been widely realized: generation of maps and generation of tabular reports.
- As spatial information is organized in a GIS, it should be able to answer complex questions regarding space.
- Spatial analysis in GIS involves three types of operations: 1.Attribute Query also known as non-spatial (or aspatial) query, 2. Spatial Query and 3. Generation of new data sets from the original database.
- The scope of spatial analysis ranges from a simple query about the spatial phenomenon to complicated combinations of attribute queries, spatial queries, and alterations of original data.

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3.1 VECTOR BASED SPATIAL DATA ANALYSIS

- There are various vector operations that dealt in multi layer operations, which allow combining features from different layers to form a new map and give new information and features that were not present in the individual maps.

1. **Topological overlays:** Selective overlay of polygons, lines and points enables the users to generate a map containing features and attributes of interest, extracted from different themes or layers. Overlay operations can be *performed on both raster (or grid) and vector maps.*

- In topological overlays polygon features of one layer can be combined with point, line and polygon features of a layer

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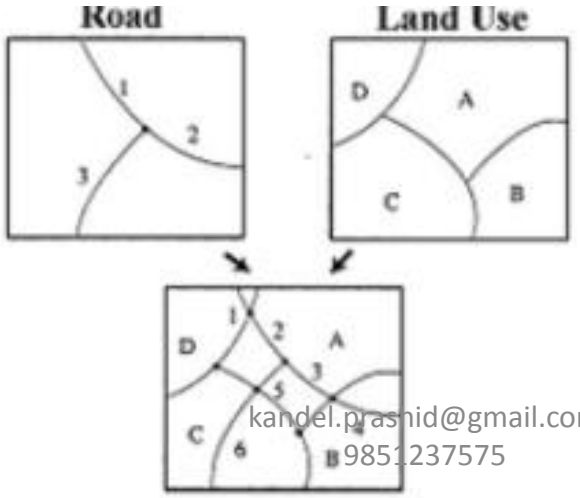
- <u>Polygon-in-polygon overlay</u>: Output is polygon coverage. Coverages are overlaid two at a time. There is no limit on the number of coverages to be combined. New File Attribute Table is created having information about each newly created feature.

- <u>Line-in-polygon overlay</u>: Output is line coverage with additional attribute. No polygon boundaries are copied. New arc-node topology is created.



Line Topology

ID	Road	Land Use
1	1	D
2	1	A
3	2	A
4	2	B
5	3	A
6	3	C

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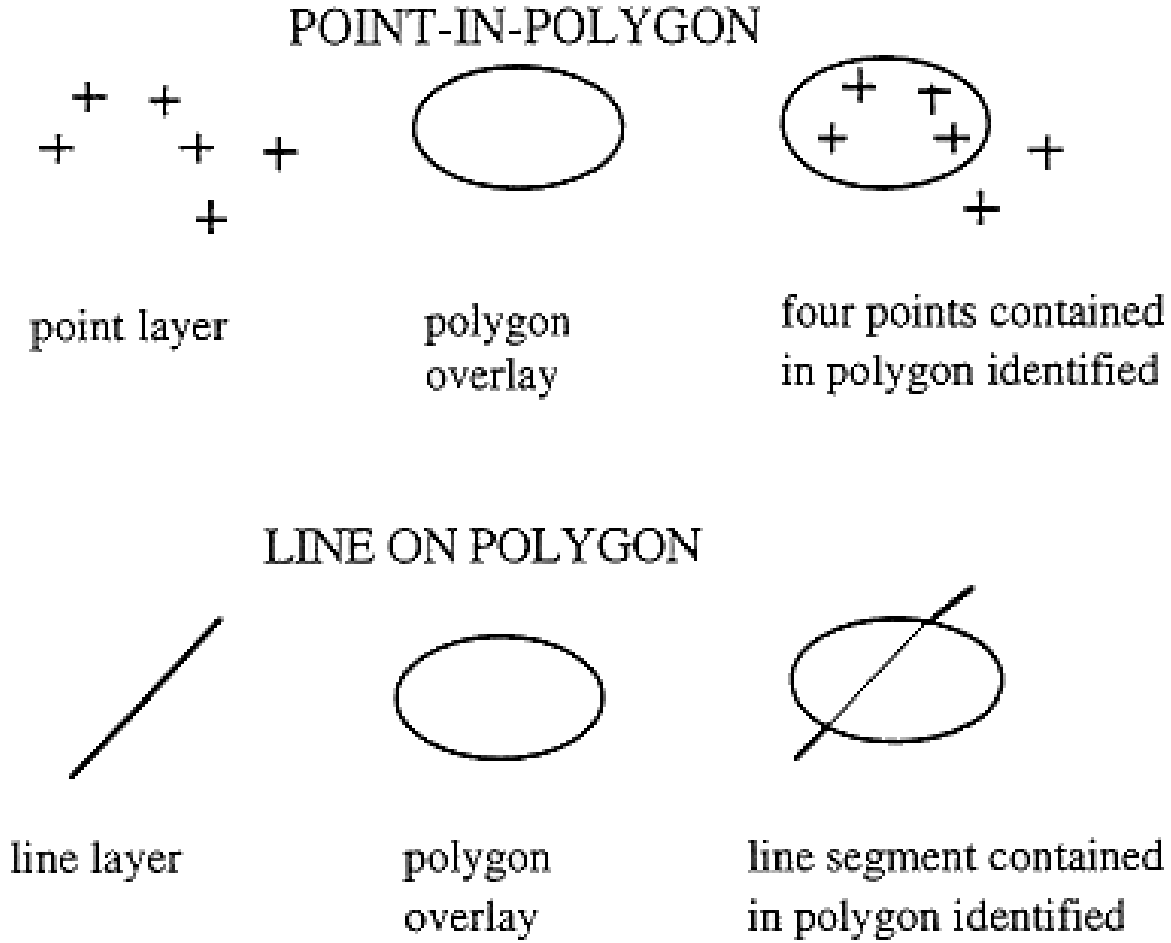
- <u>Point-in-polygon overlay:</u> Output is point coverage with additional attributes. No new point features are created. No polygon boundaries are copied.



2. Logical Operators:

- Overlay analysis manipulates spatial data organized in different layers to create combined spatial features according to logical conditions specified in Boolean algebra with the help of logical and conditional operators.
- The logical conditions are specified with operands (data elements) and operators (relationships among data elements). *Note: In vector overlay, arithmetic operations are performed with the help of logical operators. There is no direct way to it.*
- Common logical operators include AND, OR, XOR (Exclusive OR), and NOT. Each operation is characterized by specific logical checks of decision criteria to determine if a condition is true or false.

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AND - Common Area/ Intersection / Clipping Operation

OR - Union Or Addition

NOT - (Inverter)

XOR - Minus

Table 1: Truth Table of common Boolean operations

A	B	A AND B	A OR B	A NOT B	B NOT A	A XOR B
0	0	0	0	0	0	0
0	1	0	1	0	1	1
1	0	0	1	1	0	1
1	1	1	1	0	0	0

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- The most common basic multi layer operations by using Vector datasets are union, intersection, and identify operations. All three operations merge spatial features on separate data layers to create new features from the original coverage.

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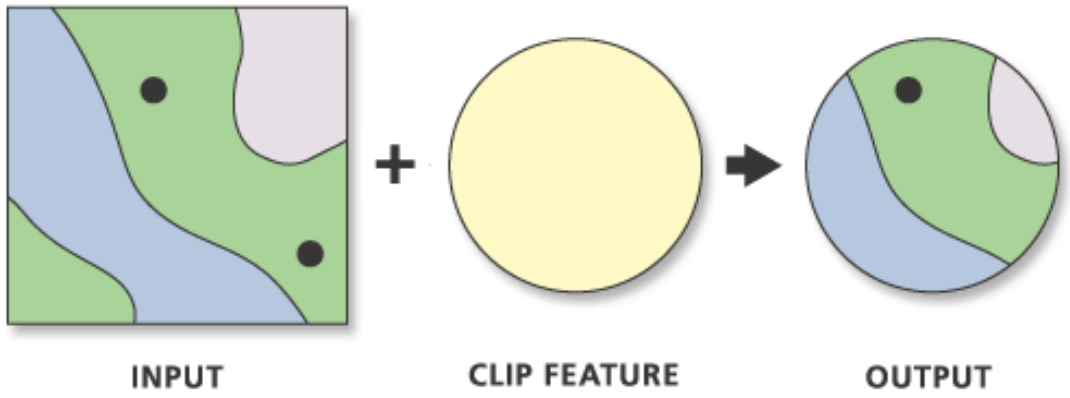
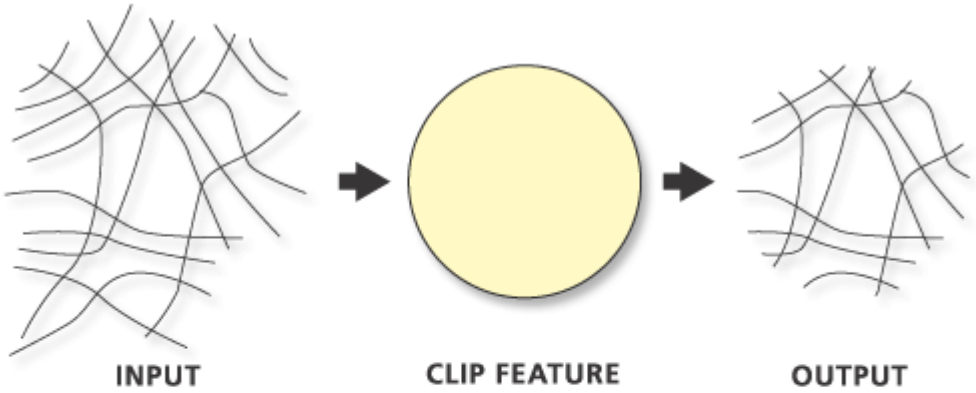
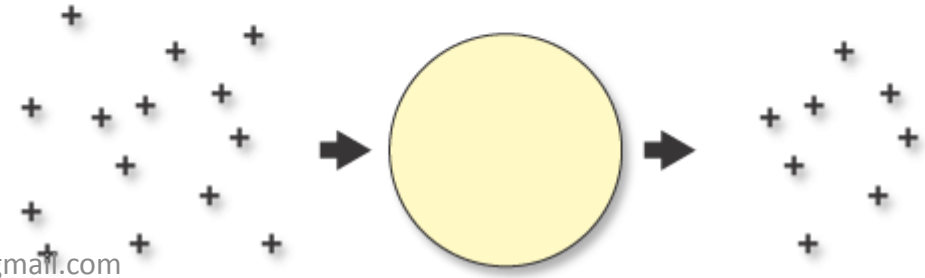
Clip: Clip creates a new coverage by overlaying two sets of features.

- The Clip Features dataset must be polygons.
- The Output Feature Class will contain all the attributes of the Input Features.





- Point features clipped by polygon features



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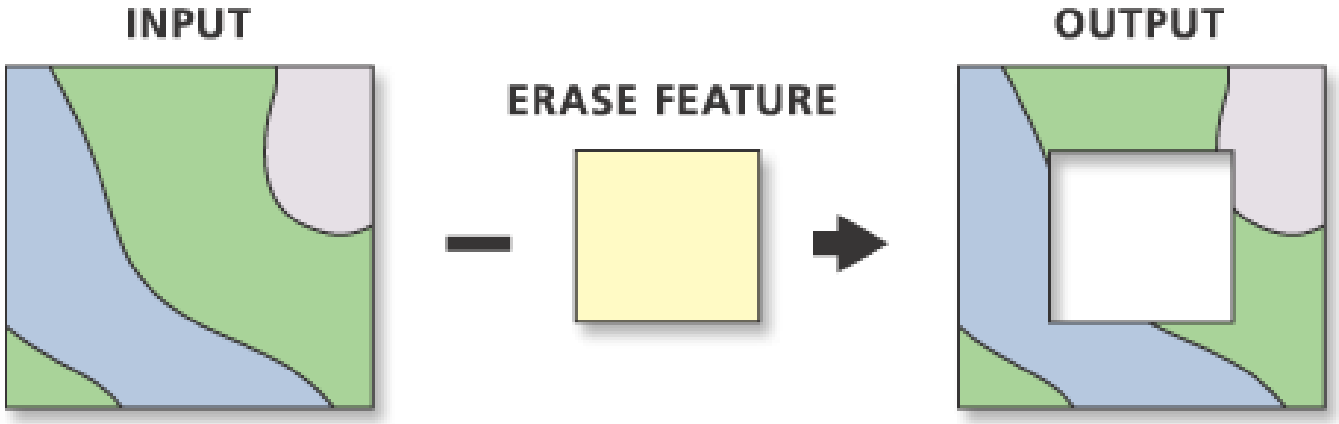
Erase: Creates a feature class by overlaying the Input Features with the polygons of the Erase Features.



- Only those portions of the input features falling outside the erase features outside boundaries are copied to the output feature class.
- The Erase Features can be point, line, or polygon.

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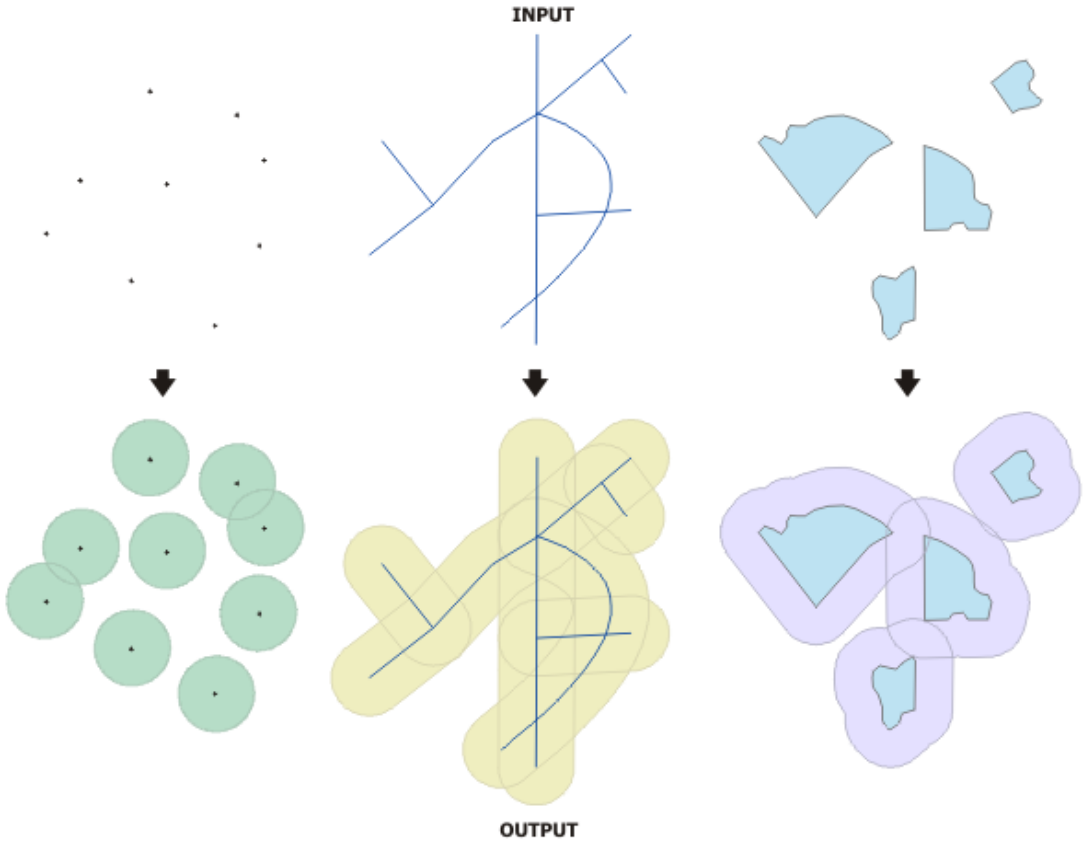
Buffer: Creates buffer polygons around input features to a specified distance.



- Buffer distances can be provided as either a value representing a linear distance or a field from the input features that contains the linear distance to buffer each feature.

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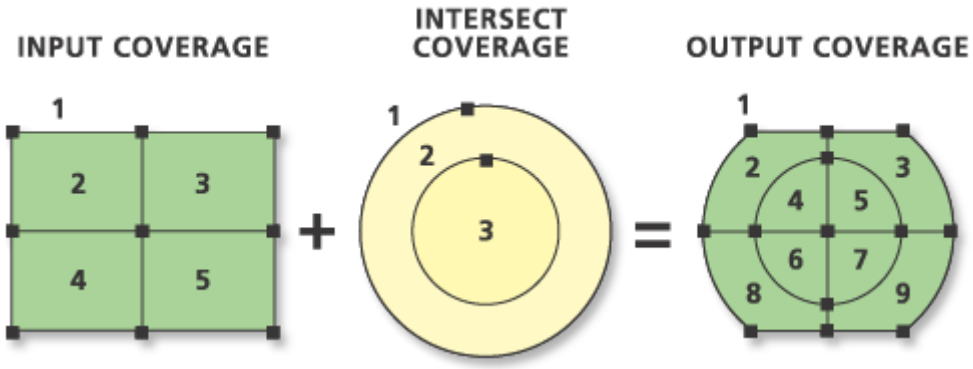
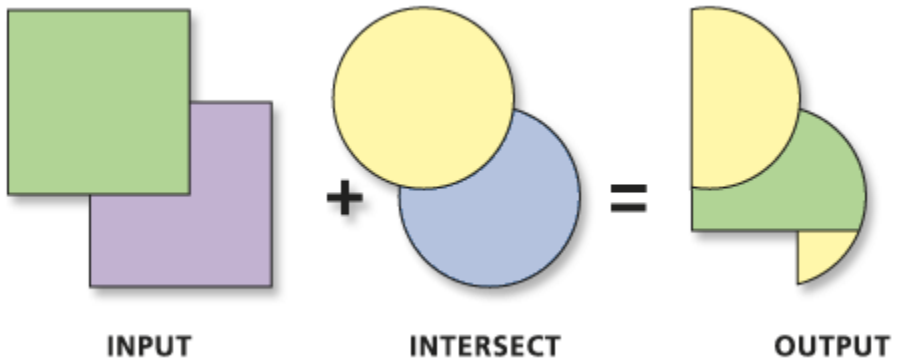
Intersect: creates a new coverage by overlaying the features from the input coverage and intersect polygon coverage.

- The output features have the attribute from the original feature from the input coverage and the feature in the intersect coverage, which they intersect.
- Intersect is one of several Overlay tools available. The tool most similar to Intersect is Clip, which does not transfer any attribute from the overlay feature class to the output.





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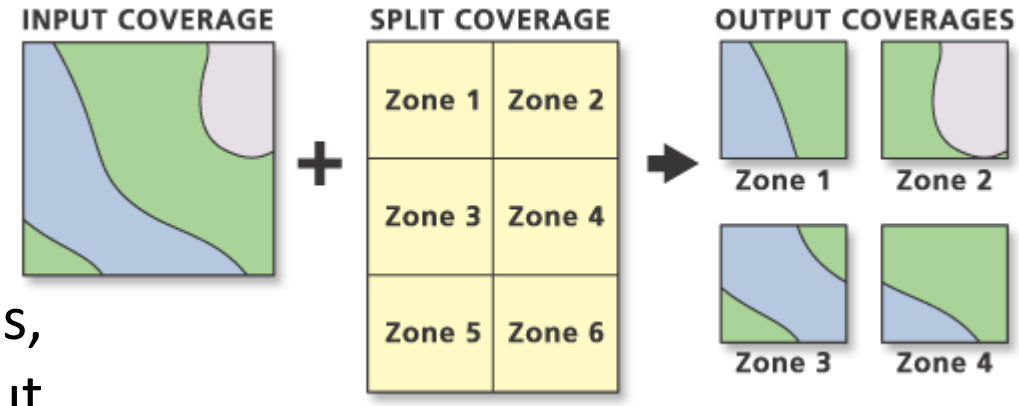
SPLIT: creates output coverages by overlaying two sets of features.



- Split performs a series of Clip operations, one for each output coverage. Each output coverage contains only those portions of input coverage features that are overlapped by the specified polygons of the Split Coverage.

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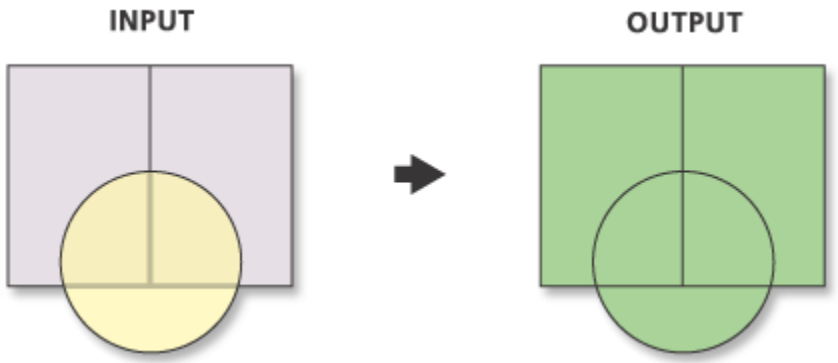
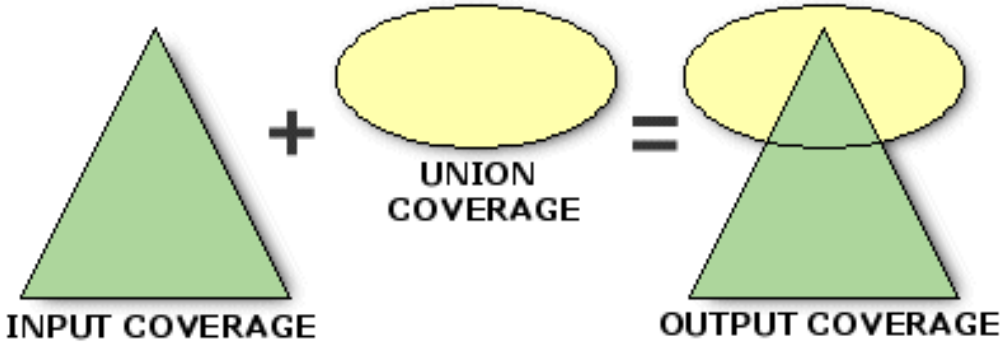
Union: Computes the geometric intersection of two polygon coverages.

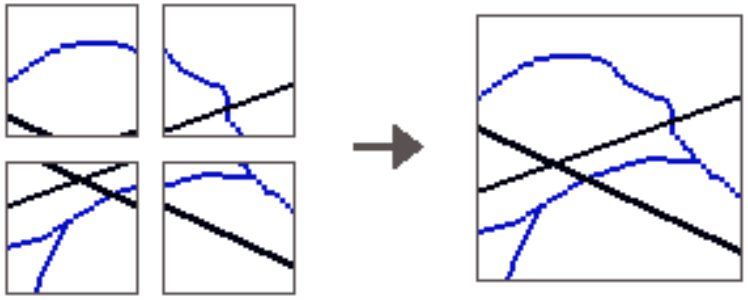


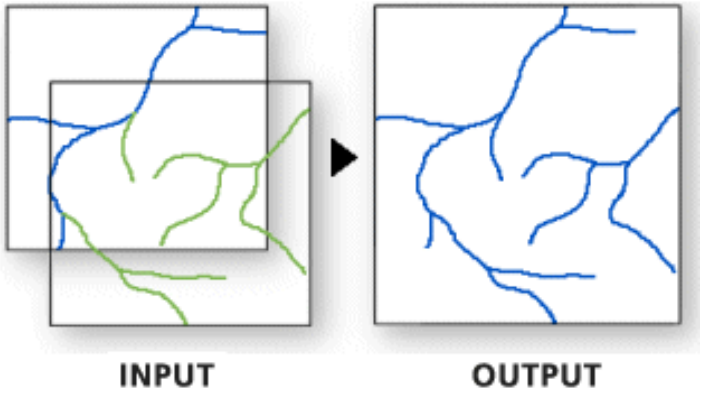


- All features will be written to the Output Feature Class with the attributes from the Input Features, which it overlaps.
- All input feature classes and feature layers must have polygon geometry.

Append: Combines an unlimited number of coverages into a single coverage.

- Input datasets can be point, line, or polygon feature classes, tables, rasters, raster catalogs, annotation feature classes, or dimensions feature classes.
- To combine input datasets into a new output dataset, use the Merge tool.





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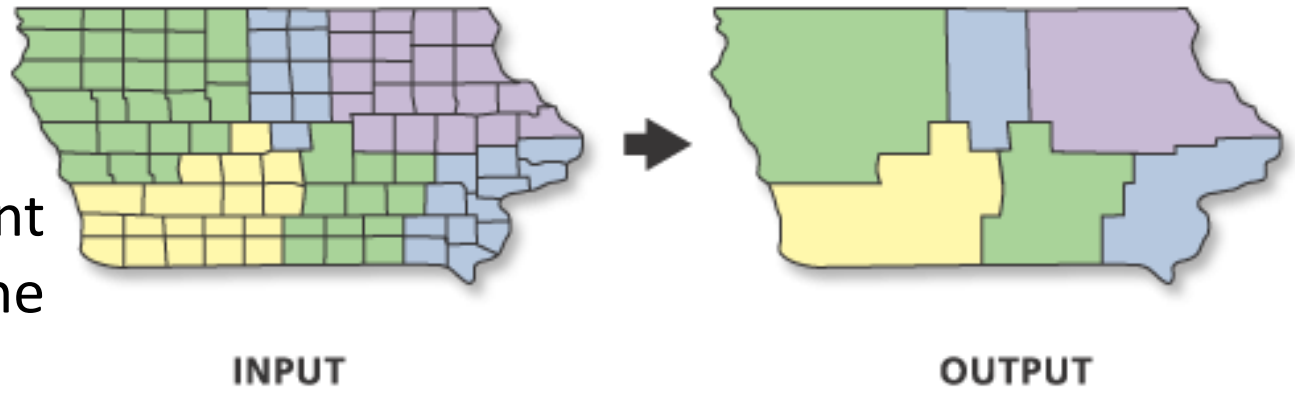
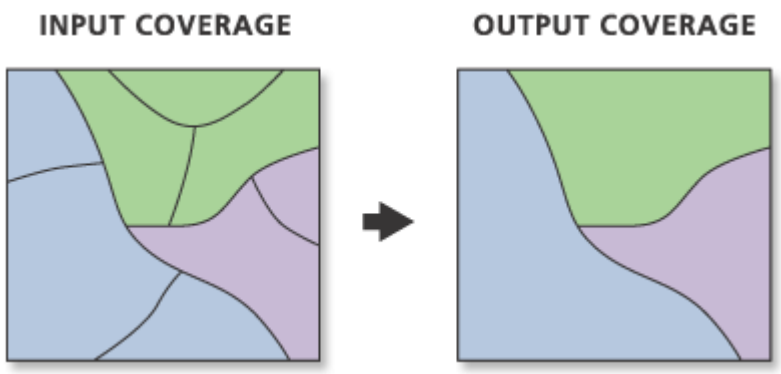
Dissolve: Aggregates features based on specified attributes.

- Creates a new coverage by merging adjacent polygons, lines, or regions that have the same value for a specified item.
- Dissolve may result in multipart features being created. A multipart feature is a single feature that contains noncontiguous elements and is represented in the attribute table as one record.





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- Similarly, there are various function perform by various tools for the spatial analysis using only vector dataset. Like, Spatial Join, Merge, extraction, Proximity.
- Each of these function could be used accordingly on the availability of the data and purpose of the analysis.

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Raster-based Spatial Analysis

- In raster analysis, geographic units are regularly spaced, and the location of each unit is referenced by row and column positions.
- Because geographic units are of equal size and identical shape, area adjustment of geographic units is unnecessary and spatial properties of geographic entities are relatively easy to trace.
- The regularity in the arrangement of geographic units allows for the underlying spatial relationships to be efficiently formulated.

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Advantages of using the Raster Format in Spatial Analysis

- Efficient processing: Because geographic units are regularly spaced with identical spatial properties, multiple layer operations can be processed very efficiently.
- Numerous existing sources: **Grids are the common format** for numerous sources of spatial information including satellite imagery, scanned aerial photos, and digital elevation models, among others. These data sources have been adopted in many GIS projects and have become the most common sources of major geographic databases.
- Different feature types organized in the same layer: For instance, the same grid may consist of point features, line features, and area features, as long as different features are assigned different values.

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Some Raster operations used in Spatial Analysis

- Those that work on single cell locations (local operations)
- Those that work on cell locations within a neighborhood (focal operations)
- Those that work on cell locations within zones (zonal operations)
- Those that work on all cells within the raster (global operations)
    - **Euclidean distance global operations**
    - **Non-Euclidean (weighted) distance global operations**
- Those that perform a specific application (for example, hydrologic analysis operations)

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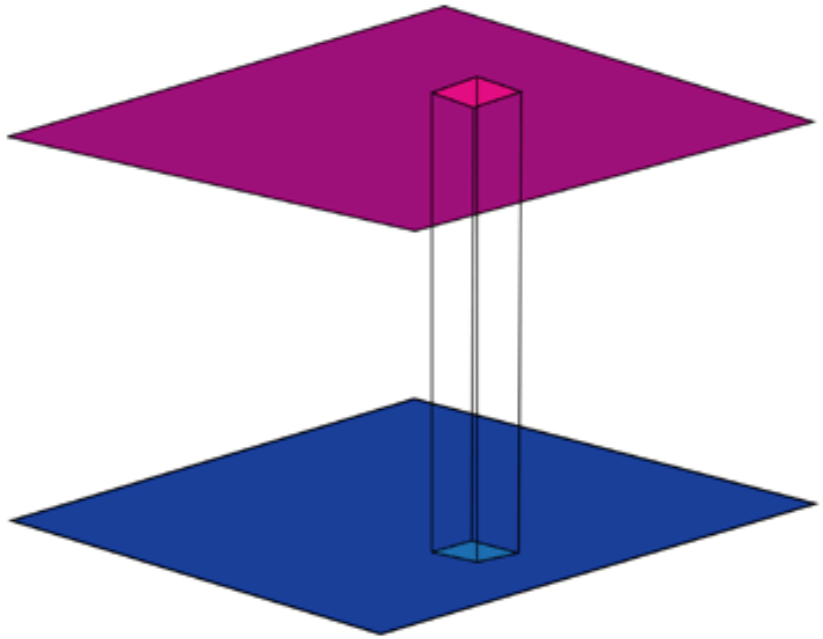
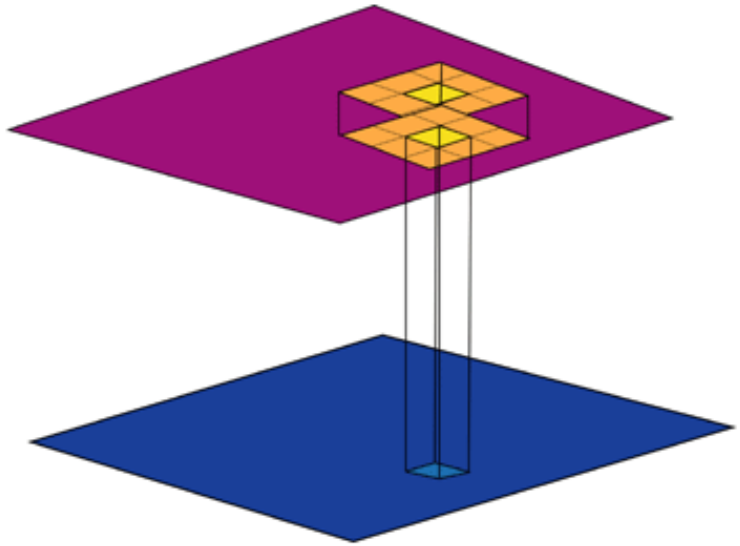
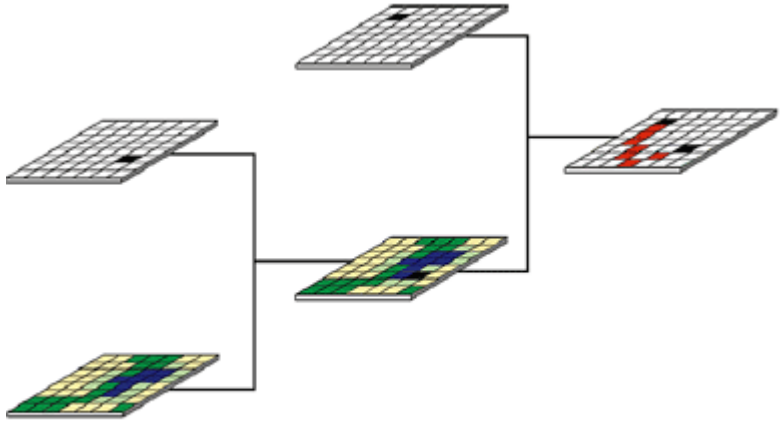


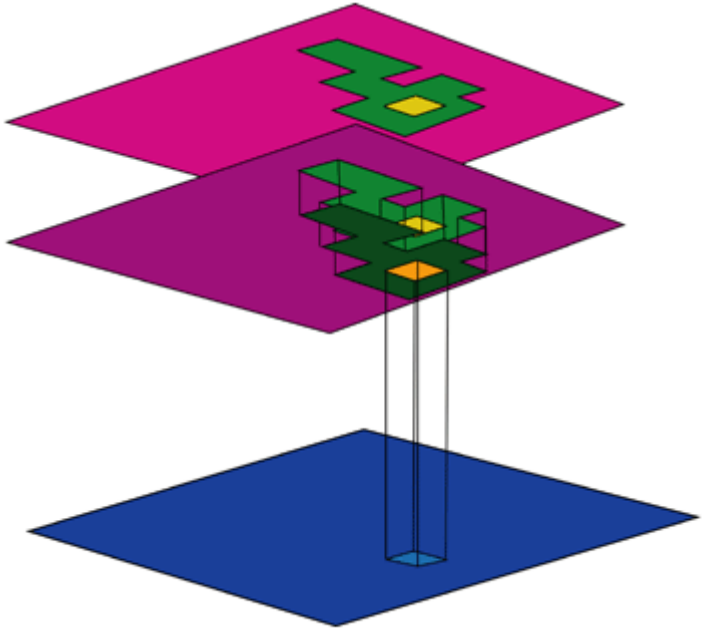
Local operations: Value of an output cell determined by a single input cell.



Focal operations: Value of the output cell determined by the cells specified neighborhood around each input cell.



An example of a global operation is Euclidean distance.



Zonal operations: Value of each output cell determined by all the input cells of the same zone.

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Application operations include the following:

- Altering raster resolution
- Density analysis
- Hydrologic analysis
- Image classification
- Generalization
- Surface analysis
- Surface generation
- Renumbering Areas in a Grid File
- Performing a Cost Surface Analysis
- Performing an Optimal Path Analysis
- Performing a Proximity Search

There are two extensions to ArcGIS that enhance the analysis capabilities of raster data. These are

•**ArcGIS Spatial Analyst**—This extension provides a comprehensive set of advanced spatial modeling and analysis tools that allow you to perform integrated raster and vector analysis.

•**ArcGIS 3D Analyst**—This extension allows you to effectively visualize, analyze, and generate surface data and provides the tools for three-dimensional modeling and analysis.

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What is Map???????

- A map is defined as a representation, usually on a flat surface, of a whole or part of an area.
- The job of a map is to describe spatial relationships of specific features that the map aims to represent.
- There are many different types of maps that attempt to represent specific things.
- Maps can display political boundaries, population, physical features, natural resources, roads, climates, elevation (topography), and economic activities.

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- Maps are produced by cartographers. Cartography refers both the study of maps and the process of map-making. It has evolved from basic drawings of maps to the use of computers and other technologies to assist in making and mass producing maps.

There are four basic kinds of maps

Physical

Transportation

Political

Historical

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IS A GLOBE A MAP?

A globe is a map. Globes are some of the most accurate maps that exist.

This is because the earth is a three-dimensional object that is close to spherical.

A globe is an accurate representation of the spherical shape of the world.

Maps lose their accuracy because they are actually **projections** of a part of or the entire Earth.

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Elements of Maps

- See it in Unit 2 which is already mentioned there.
- Just try to remember at least 10 of those important elements

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UNIT 6 – INTRODUCTION TO SPATIAL DATA STRUCTURE

- SDI concepts and its current trend
- The concept of metadata and clearing house
- Critical factors around SDIs

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SDI (Spatial Data Infrastructure)

- The term spatial data infrastructure was coined in 1993 by the U.S. National Research Council to denote a framework of technologies, policies, and institutional arrangements that together facilitate the creation, exchange, and use of geospatial data and related information resources across an information-sharing community.
- Such a framework can be implemented narrowly to enable the sharing of geospatial information within an organization or more broadly for use at a national, regional, or global level.

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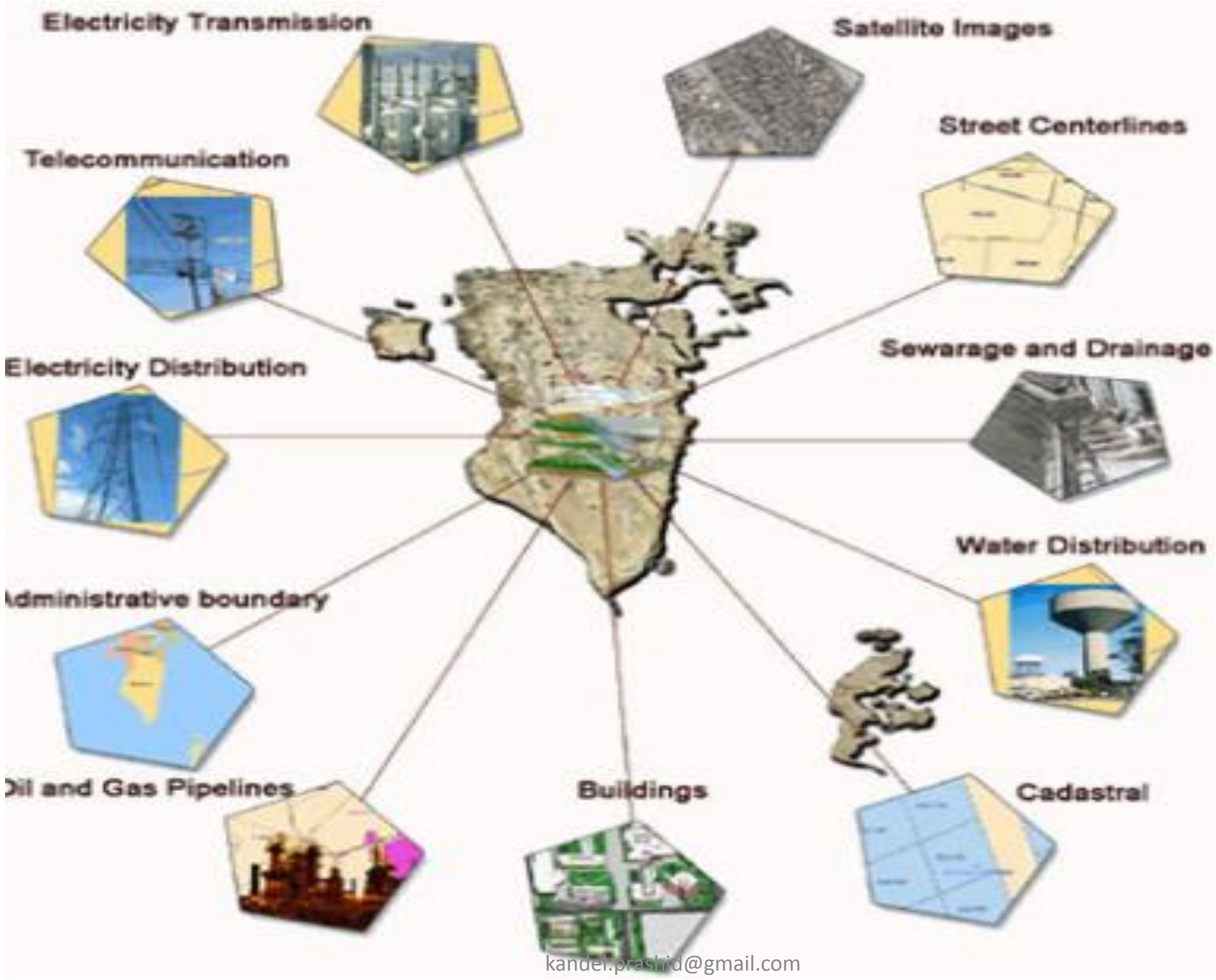

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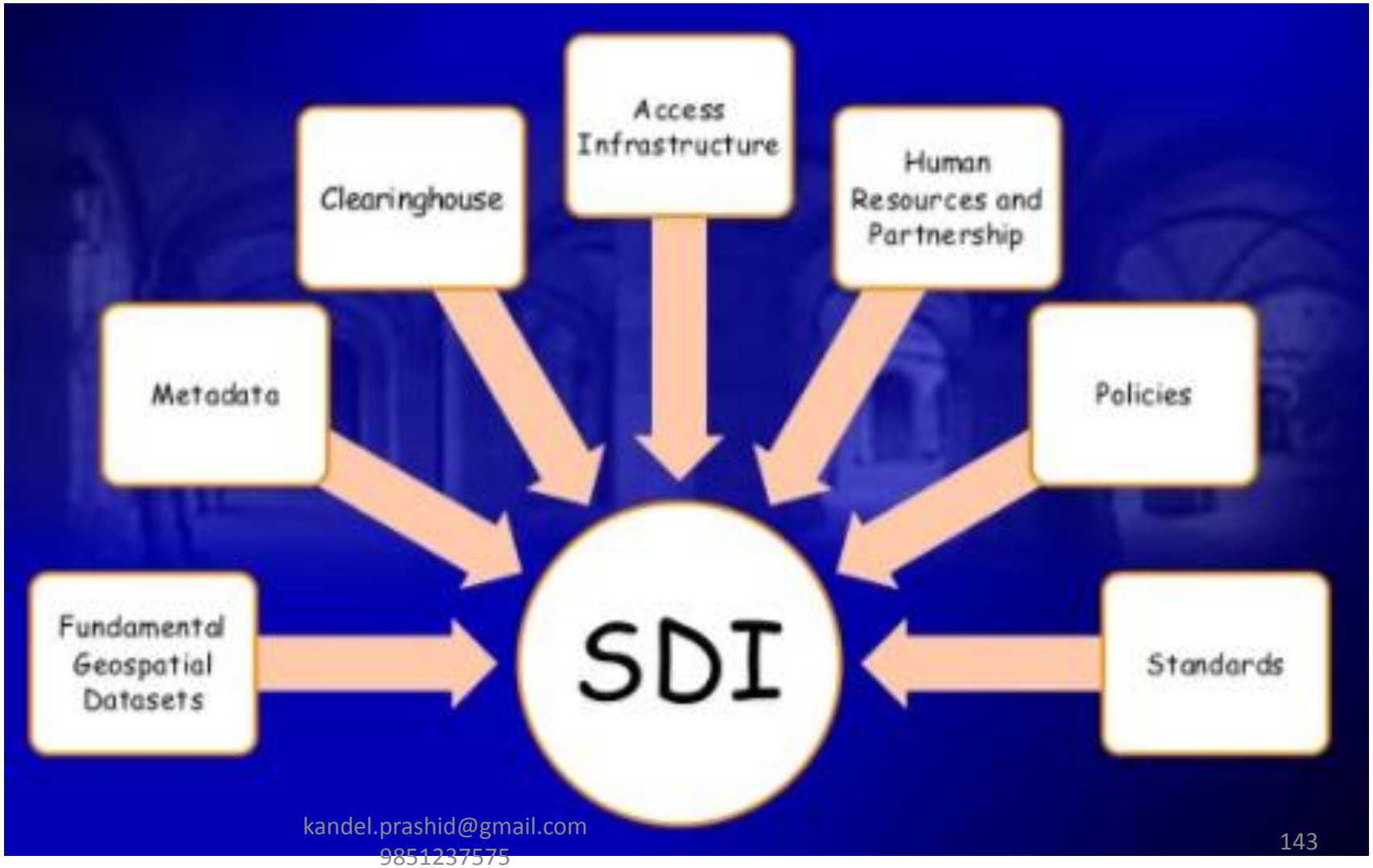
- In all cases, an SDI will provide an institutionally sanctioned, automated means for posting, discovering, evaluating, and exchanging geospatial information by participating information producers and users.
- SDI extends a GIS by ensuring geospatial data and standards are used to create authoritative datasets and polices that support it.
- It is an initiative intended to create an environment that enables a wide variety of users to access and retrieve complete consistent data sets in easy and secure way.
- It is defined as the technology, policies, standards, and institutional arrangements necessary to acquire, process, store, distribute and improve the utilization of geospatial data from different sources and for a wide range of potential users.

Components of SDI



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Benefits of SDI

- **Reduce time and redundancy of data production**- By connecting geospatial data and service producers and consumers, infrastructure and costs can be kept economical. The technology is scalable and can fit into any infrastructure, including existing geospatial technology installations.
- **Enable easy search and discovery of existing geospatial data and services** - Users can create and post metadata records efficiently, which are necessary for inventorying, locating, and assessing the quality of geospatial data.
- **Maintain data integrity and security**- Organizations easily share the authoritative version of data among users and provide mechanisms for the creation of authentic data and metadata.

Clearinghouse

- A (spatial data) clearinghouse is a distributed network of spatial data producers, managers and users that are linked electronically together.
- It is a system of software and institutions that are to facilitate the discovery, evolution and downloading of digital spatial data and provides means to inventory, document and data sharing.
- The clearinghouse concept is a useful one in building a Geographic Information Infrastructure (GII).
- The objective is to minimize unnecessary duplication of effort for data capture, and to maximize the benefit of geographic information sharing.
- Data providers nowadays are fully aware of the importance of advertising and making available their metadata describing their databases, to facilitate the use of their products. This explains the current level of activity of building these clearinghouses.

Data Clearinghouse!!

- Institutional view
  People and infrastructure to facilitate discovery of who has what geographic information.

- Technical view
  A set of information services that use hardware, software, and telecommunications networks provide searchable access to information.

How does a clearinghouse work?

- A clearinghouse allows data providers to register their geographic data sets, the quality of these data and also the instructions for accessing them. Each data provider provides an electronic description of each spatial data set. In addition, the provider may also provide access to the spatial data set itself. The clearinghouse thus functions as a detailed catalogue service with support for links to spatial data and browsing capabilities.

Metadata

- Metadata is defined as background information that describes the content, quality, condition and other appropriate characteristics of the data.
- So metadata is a simple mechanism to inform others of the existence of the data sets, their purpose and scope.
- In essence metadata answer who, what, when, where, why and how questions (WH Questions) about all facets of the data made available.
- Metadata can be used internally be the data provider to monitor the status of data sets, and externally to advertise to potential users through a national clearinghouse.
- Metadata are important in the production of a digital spatial data clearinghouse, where potential users can search for the data they need.

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Roles of Metadata

- Applicability: information needed to determine the data sets that exists for a geographic location
- Fitness for use: information needed to determine whether a data set meets a specified need
- Access: information needed to acquire an identified data set
- Transfer: information needed to process and use a data set
- Administration: information needed to document the status of existing data (data model, quality, completeness, temporal validity etc…) to define internal policy for update operations from different data sources.

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Metadata Standards

- For metadata to be easily read and understood, standards create a common language for users and producers.
- Metadata standards provide appropriate and adequate information for the design of metadata.
- The key development in metadata standards are the ISO STANDARD 1504615 METADATA, the federal geographic data committee's content standard for Digital Geospatial Metadata FGDC.
- A standard provides a common terminology and definitions for the documentation of spatial data.

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Critical Factors on SDI

- Data sharing problems which can be viewed as critical factors in SDI can be briefly described as follows:
  - **Data standards**: It refers to an agreed upon way of representing data in a system in terms of content, type and format. Exchange of data between databases is difficult it they support different data standards or different query language. Examples of these standards are the DIGEST (Digital Geographic Information Exchange Standard) Spatial Data Transfer Standard (SDTS) etc.
  - **Heterogeneity**: it means being different in kind, quality and character. Spatial data may exists in a variety of locations, are possibly managed by a variety of database systems, were collected for different purposes and by different methods, and are stored in different structures. This brings about all kinds of inconsistency among these data sets (heterogeneity) and creates many problems when data is shared.

- **Communication problems**: with advances in computer network communication and related technology, locating relevant information in a network of distributed information sources has become more important recently. The question is which communication technology is the best suitable for transfer of huge amounts of spatial data in a secure and reliable way. Efficient tools and communication protocols are necessary to provide search browse and delivery mechanisms.
- **Institutional and Economic problems**: these problems arise in the absence of policy concerning pricing, copyright, privacy, liability, conformity with standards, data quality etc… resolving these problems is essential to create the right environment for data sharing

UNIT 7 – OPEN GIS

- Introduction of open concept in GIS
- Open source software for spatial data analysis
- Web Based GIS system
- System Analysis and Design with GIS

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OPEN CONCEPT IN GIS

- Open source is a software concept of allowing the source code to be open and free to distribute and modify.
- There is "The Open Geospatial Consortium, Inc (OGC) which is an international industry consortium of **368** companies, government agencies and universities participating in a consensus process to develop publicly available interface specifications. OpenGIS Specifications support interoperable solutions that "geo-enable" the Web, wireless and location-based services, and mainstream IT. The specifications empower technology developers to make complex spatial information and services accessible and useful with all kinds of applications.

OGC

- The OGC (Open Geospatial Consortium) is an international not for profit organization committed to making quality open standards for the global geospatial community. These standards are made through a consensus process and are freely available for anyone to use to improve sharing of the world's geospatial data.
- OGC standards are used in a wide variety of domains including Environment, Defense, Health, Agriculture, Meteorology, Sustainable Development and many more.

- MAGIC uses PostgreSQL database and PostGIS spatial layers in its online mapping tools and creates geospatial data that meet OGC standards.Data created in MAGIC are accessible in Open Source GIS software.Other open source GIS applications include the GDAL raster plugin, Quantum GIS, Grass and many others. http://www.opengeospatial.org/

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Open source software for spatial data analysis

- QGIS
- **gVSIG**
- **Whitebox GAT**
- **SAGA (System for Automated Geoscientific Analyses) GIS**
- **GRASS (Geographic Resources Analysis Support System) GIS**
- **Mapwindow**
- **ILWIS (Integrated Land and Water Information Management)**
- **GeoDa**
- **uDig**
- **OpenJUMP**
- **Diva GIS**
- **Falcon View**
- **OrbisGIS**

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Web Based GIS system

- **Web-based GIS** is based on a type of distributed information. This set of technological services is part of a communication structure between the **GIS** server and the client.
- Their relationship is expressed through URLs (created by the server) and HTTP (for the customer). Spatial data access, advanced mapping and spatial analysis are the most common type of analysis options in **Web-based GIS.**
- Access to spatial data as well as advanced mapping and spatial analysis over the Internet is becoming more common.

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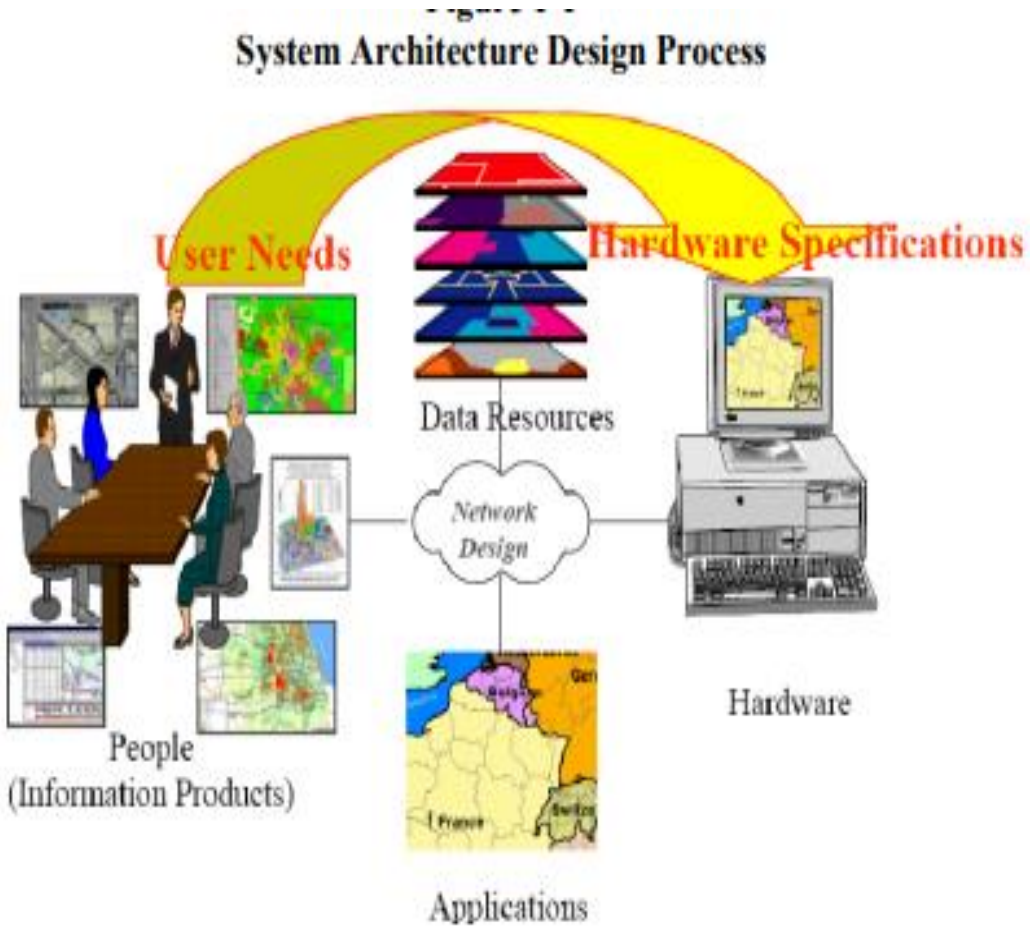
Some of the commonly used Web Map Services:

- **ArcGIS Server**
  Documentation, tutorials, and tips and tricks for using ArcGIS Server from ESRI. With this previous generation online map serving technology, you can create simple, “out-of-the box” map services with pan, zoom and simple query functions.
- **AspMap**
  AspMap is a Web mapping component for embedding spatial data access, display and analysis capabilities in Web applications and services. AspMap supports ASP and ASP.NET.

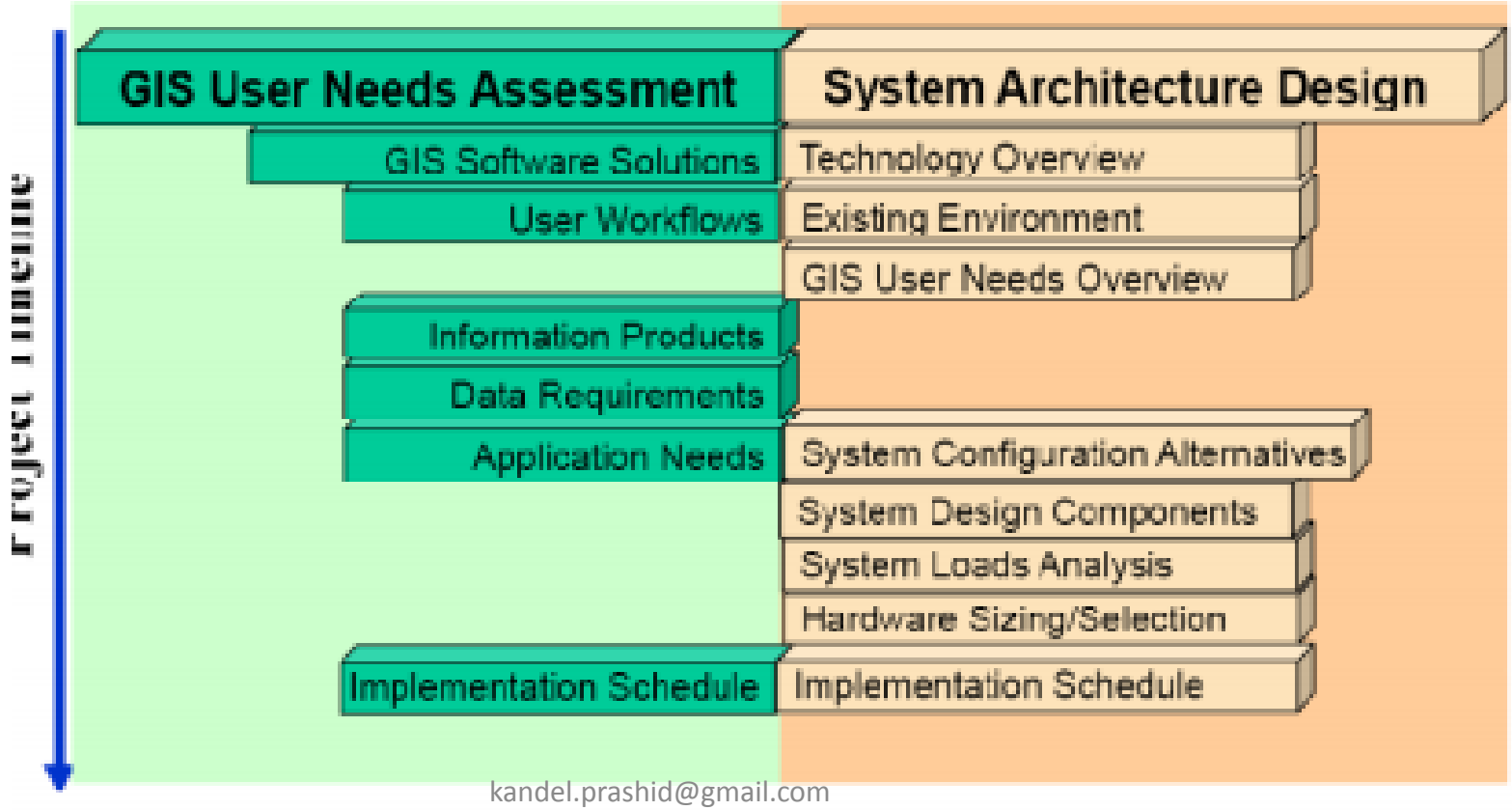
- **Google Earth**
  Google Earth combines satellite imagery, geographic data and Google's search capabilities to create a virtual globe application that you can download to your desktop to access online spatial data. Find resources here to help you expand this geographic data viewer application.
- **Google Maps**
  Using Google Maps on individual web sites, called map hacking, is a fast and easy way to add geographic information to personal web sites.
- **Mapserver**
  MapServer is an OpenSource development environment for building spatially enabled Internet applications. The software builds upon other popular OpenSource or freeware systems like Shapelib, FreeType, Proj.4, libTIFF, Perl and others.
- **Map-TV**
  Map-TV is a mapserver that has a built in web server. This map server uses ESRI shapefile format for data display.

System Analysis and Design with GIS

- System architecture design is a process developed by ESRI to promote successful GIS implementations. This process supports existing infrastructure requirements and provides specific recommendations for hardware and network solutions based on existing and projected user needs. Application requirements, data resources, and people within an organization are all important in determining the optimum hardware solution as shown in side figure.



- The system architecture design is based on user workflow requirements identified in the GIS needs assessment. The most effective system design approach considers user needs and system architecture constraints throughout the design process.



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- A GIS professional consultant familiar with current GIS solutions and customer business practices can help facilitate this planning effort.
- The system design begins with technology exchange. Technology exchange provides the foundation for client support throughout the design process. Client participation is a key ingredient in the design process. The design process includes a review of the existing computer environment, GIS user requirements, and system design alternatives.
- The system design models translate peak capacity hardware loads developed during the user workflow analysis to platform specifications to support operational requirements.